



TECHNICAL REFERENCE

Power Glove Vision Configuration Reference

Active files, installed copies, fields, secrets, and generated state.

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Use this reference to find a setting, change a game mapping, tune a gesture, or look up a command. Each section identifies the active file and explains what its values mean.

Use the [installation guide](#) for the initial deployment and pairing procedure. Return here when you need to change a host, camera, game, gesture threshold, or network setting.

KEEP THE TOKEN PRIVATE PowerGlove Vision uses one shared token to authenticate controller packets and profile changes. Never paste it into an issue, screenshot, command line, public backup, or Git commit.

The three places configuration lives

PowerGlove Vision runs on two computers. A repository template is not always the file the running system reads.

Location	What it controls	Preferred way to change it
UNO Q Setup page	Console address, controller port, startup profile, camera, and pairing token	Browser Setup page
UNO Q application files	Gesture sensitivity and advanced runtime defaults	Edit only when tuning is required
RetroPie <code>/etc/powerglove/</code>	UNO Q address, per-game profile selection, and receiver token	Protected files on RetroPie

The examples under the repository's `config/` directory are installation templates. Editing them does not change an already installed system. The active copies are identified in each section below.

In commands and examples, replace these placeholders:

Placeholder	Replace with
<code>UNO-Q-NAME.local</code>	Your UNO Q hostname or reserved IP address
<code>RETROPIE-NAME.local</code>	Your RetroPie hostname or reserved IP address
<code>/home/arduino/ArduinoApps/powerglove-vision</code>	Required directory for the supported UNO Q installer and shutdown helper; do not substitute a different path

Find the setting or command you need

Task	Section
Install and pair both machines	Installation Guide
Change the camera or startup profile	UNO Q settings
Repair pairing or token permissions	Pairing and token management
Change the UNO Q destination on RetroPie	RetroPie connection settings
Make a game select a profile	Register games and select profiles
Adjust gesture sensitivity	Tune gesture sensitivity
Understand an option or command	Command-line reference

UNO Q settings

Open the ordinary Setup page at:

```
http://UNO-Q-NAME.local:8088/setup
```

Pairing uses the secure Setup page instead:

```
https://UNO-Q-NAME.local:8443/setup
```

The secure page uses a locally generated certificate. During pairing, compare the certificate fingerprint shown by the browser with the identifier shown on the UNO Q matrix before entering the one-time PIN.

Settings shown in the browser

Setting	Default	Meaning and recommendation
Console hostname or IP	Empty (not configured)	Set your RetroPie hostname (<code>RETROPIE-NAME.local</code> in examples) or a reserved LAN address and pair through Connection before starting controls. Glove Academy and local settings work without a destination. Existing saved destinations are preserved.
Controller port	55355	UNO Q to RetroPie controller-state port. Leave it at the default unless both ends are changed.
Startup profile	bad_street_brawler	Profile used before a registered game selects another one.
Tracking aid	none	none, white, or black. In the current release this is an informational diagnostic label; it does not change MediaPipe tracking.
Camera	auto	Prefer auto. Use a number from 0 through 99 only when automatic selection chooses the wrong capture device.
Generate a new token	Off	Rotates the shared secret. This immediately breaks the existing pairing until RetroPie is paired again.

Selecting **Save** validates the fields, writes them atomically with private permissions, and restarts the vision worker using the saved calibration. Recalibrate only if you have moved the camera, changed your playing position, or notice unwanted movement while your hand is at rest.

The Dashboard profile selector changes only the current active profile. It does not rewrite `device.json` or change the Setup page's startup profile. RetroPie may replace a Dashboard selection when a game starts or ends.

Vision startup and timing

Camera capture uses the complete 640×480 field of view at 60 fps. A dedicated capture thread continuously drains the camera and retains only its newest frame, so inference skips superseded images instead of building an input queue. The deployed worker explicitly selects **MediaPipe Hands (proven)**, whose stable command identifier is `legacy`, and lets each of its two inference stages use up to four CPU threads (`--inference-threads 4`). Thread-count benchmarking remains part of the 0.3.2 performance work; more threads are not assumed to be faster on every device. **MediaPipe Tasks Video (experimental)** remains available under the stable `tasks-video` identifier for controlled comparison.

Camera-preview drawing and JPEG encoding run only while a browser is actively watching the Dashboard or Glove Academy stream. JPEG encoding runs on a separate latest-preview worker and is allowed to drop superseded preview jobs rather than delay another controller sample. Closing those pages avoids even that optional work. When a preview is open, submissions are limited to 5 fps (`--preview-fps 5`).

The worker preloads OpenCV and MediaPipe on its background vision thread as soon as its control server is available. Preloading imports the libraries; it does not open the camera, create a hand tracker, or process images. The website and profile controls remain responsive while imports run. A failed preload is logged; a later activation retries loading and reports any error.

If **Gestures off** is selected at startup, the camera stays closed until you select an active profile or open Glove Academy. An active startup profile requests capture automatically after preloading. Controller delivery still starts stopped.

Activation waits for any unfinished preload, verifies the saved model, opens and configures the camera, waits for a usable frame, and creates the tracker. Dashboard and Glove Academy show **Starting camera and gesture tracking** until vision is active. The elapsed time covers startup work, not only the physical camera. **Calibrate** stays disabled until initialization finishes.

Switching between active profiles reuses the camera and tracker. **Gestures off** releases both, while imported libraries remain in memory. An application restart, including a worker restart after saving Setup settings, starts preloading again.

Measurements on the cabinet's UNO Q with a Razer Kiyo Pro on September 4, 2026:

Measurement	Observed time
First activation after reboot, before background preloading	7.67 seconds
OpenCV background preload after a later reboot	0.94 seconds
MediaPipe background preload after that reboot	5.94 seconds
First activation after that preload completed	1.21 seconds

These were separate reboot tests on this cabinet, not guaranteed timings. The change moves library loading earlier; it does not eliminate that work. Selecting a profile immediately after application startup can still wait for preloading. An earlier reported 13-14 second delay was not reproduced in the instrumented tests.

To inspect startup stages on the UNO Q:

```
SH
docker logs --since 10m powerglove-vision-main-1 2>&1 | grep 'Vision startup:'
```

The default installation uses this container name; use `docker ps` to find it if your App Lab installation uses another name. Logs include background import times, model verification/recovery, camera discovery/open/settings, the first camera frame, tracker construction, first inference, and time to active worker status. Preparation and activation totals include earlier stages; do not add them to the individual durations. Dashboard polling adds a small delay before it displays the new status. Compare entries from the same activation.

If the camera is unavailable, check `lsusb` and `/dev/v4l/by-id/` on the UNO Q. The built-in `qcom-venus-encoder` and `qcom-venus-decoder` video nodes are not webcams. A camera missing from the USB device list is below MediaPipe and OpenCV; preloading cannot resolve that condition. The installed host helper waits for a sustained outage before making one guarded reset of the camera's last observed parent hub. It disables autosuspend whenever the single UVC camera is present. If that one attempt does not restore USB enumeration, check the powered hub, cable, and camera connection.

Glove Academy, calibration, and live readings

Glove Academy is the renamed Learn section. Existing `/learn` bookmarks still work. The matrix continues to show **L** for lessons and **T** for tuning.

Glove Academy starts the camera even when **Gestures off** is selected and uses a mapping-independent practice profile. It pauses controller delivery and restores the selected profile when you leave. With several Glove Academy tabs open, practice remains active until the last tab closes or its lease expires. A six-second lease timeout handles an unexpected browser close. Loading Dashboard also clears a stale session; reload Glove Academy if you want to begin practice again.

The sixteen lessons include A (index curl), B (thumb curl), Glove Zap (forward push), Pull Back, Start, Select, roll left, roll right, close hand, and menu guard. Menu guard requires curled thumb/ring with index, middle, and pinky extended and suppresses movement, A, B, Start, and Select. Completing every lesson earns Glove Master; skipped lessons must be revisited. **Start again** clears session progress. The practice indicators do not change a game's gesture mapping.

Recognition pose	See it	Required hand shape
Closed hand		Thumb and every finger curled into a comfortable fist.
Menu guard		Thumb and ring curled; index, middle, and pinky extended.

Reading or control	Meaning
Finger curl	Glove Academy shows values from 0 to 1; Dashboard uses a compact 0-to-3 display. Default ordinary curl actions engage at 0.50 and release below 0.35; saved personal pairs override these values.
V sign	Without personal adjustments, index and middle curl must be below 0.28; ring and little curl must exceed 0.42. Hold steadily for 0.50 seconds to send Start. A non-V pose must then remain visible for 0.30 seconds before Start can rearm.
Thumbs-up	Without personal adjustments, thumb curl must be below 0.32 and all four finger curls above 0.42. Hold for 0.15 seconds to send Select.
Live hand measurements	Shows curl values, thresholds, enlarged landmarks, and forward or backward movement relative to the calibrated hand size.
Calibrate	Replaces the saved resting reference. The button turns red while sampling, then blue with a brief completion message.
<code>tracker_backend</code>	Stable command identifier: <code>legacy</code> means MediaPipe Hands (proven) ; <code>tasks-video</code> means MediaPipe Tasks Video (experimental) .
<code>capture_age_ms</code>	Time from completion of the newest camera read to the beginning of inference. This is a freshness diagnostic, not the camera exposure timestamp.
<code>capture_interval_ms</code> , <code>capture_skipped_total</code>	Spacing between processed camera frames and the cumulative number intentionally superseded by newer frames.
<code>inference_ms</code> , <code>inference_interval_ms</code> , and <code>inference_hz</code>	Per-frame tracking calculation, spacing between recognition passes, and its reciprocal rate.
<code>sample_age_ms</code>	Time from completion of the newest camera read through inference and the local UDP send attempt. This is the primary UNO Q software-stage gameplay measurement.
<code>controller_transition_age_ms</code>	<code>sample_age_ms</code> recorded only when a successfully transmitted gameplay-visible controller state changes. Native X/Y changes are included.
<code>performance</code>	Rolling latest, p50, p95, and maximum values over the most recent 300 valid samples for capture age, processed-frame spacing, inference, inference spacing, send time, complete sample age, and changed-control age.
<code>preview_encode_ms</code> , <code>preview_dropped</code>	Background JPEG cost and previews discarded to protect controller responsiveness.
<code>send_ms</code>	Local controller-state send time. None of these readings alone measures the full delay from physical motion to the displayed game frame.

Finger recognition uses the strongest joint bend, including the base knuckle; a middle-knuckle bend alone can qualify. Thumb recognition uses the stronger of its two outer joints. MediaPipe 3D world landmarks are preferred. The fallback uses normalized depth with image aspect-ratio correction; palm movement still uses image coordinates. The legacy Arduino landmark bridge keeps its 2D interpretation because its depth units differ.

Glove Academy and gameplay share held finger and movement states. Glove Zap and Pull Back need two consecutive beyond-threshold observations plus 0.10 normalized palm-scale movement in the intended direction within 250 ms. Once confirmed, they remain recognized until movement falls below their respective release thresholds, and a confirmed menu pose still satisfies its lesson after the short controller pulse ends. The browser preview is capped at 5 fps; status updates follow each tracking calculation.

The app reuses its saved resting reference across Glove Academy, gameplay, profile changes, and restarts. It calibrates automatically only when that reference is missing or invalid. Use **Calibrate** after moving the camera or changing your playing position. Keep your palm near the resting position when practising finger curls so unintended movement does not obscure the finger readings. See [Saved neutral-hand calibration](#) for storage and recovery details.

Start controller and **Stop controller** affect live delivery only. After an application or system restart, delivery remains stopped until you start it. When RetroPie announces a registered game, a six-second launch guard sends no controller packets while the runcommand screen is active. Delivery resumes automatically afterward if it was already started; the guard does not change the saved profile or turn a stopped controller on.

Active UNO Q device file

The Setup page maintains this private file inside the application:

```
/home/arduino/ArduinoApps/powerglove-vision/data/device.json
```

A typical device configuration file contains the following fields:

```
JSON
{
  "receiver": "RETROPIE-NAME.local",
  "port": 55355,
  "token": "private-random-value-created-by-the-application",
  "profile": "bad_street_brawler",
  "glove_color": "none",
  "camera": "auto"
}
```

Use the Setup page for routine changes. If you must edit the JSON directly, stop the application first, keep the token unchanged, validate the file, and restore mode **0600**. JSON does not allow comments or trailing commas.

```
SH
python3 -m json.tool data/device.json >/dev/null
chmod 0600 data/device.json
```

Deleting `device.json` causes PowerGlove Vision to create a new token and first-run defaults. You must then pair RetroPie again.

Camera selection

`auto` searches stable `/dev/v4l/by-id/` capture-device links first, ignores known codec-only video nodes, and then considers ordinary `/dev/video*` capture devices. This is the most reliable choice when USB enumeration changes after a reboot.

Use an explicit camera number only for troubleshooting. For example, `0` selects `/dev/video0`, but Linux may assign that number to a different device after hardware is reconnected. Keep the camera on a powered USB hub when the UNO Q cannot supply stable power by itself.

Supported startup profiles

The valid startup profile identifiers are:

```
off
bad_street_brawler
super_glove_ball
program_a program_b program_c program_d program_e
program_f program_g program_h program_i
```

The startup profile does not assign a profile to a ROM. Per-game selection is controlled by the RetroPie game registry described below.

Pairing and token management

Both machines must hold the same token:

Machine	Active private file
UNO Q	Application <code>data/device.json</code> , in the <code>token</code> field
RetroPie	<code>/etc/powerglove/token</code>

Use one-time-code pairing whenever possible:

```
SH
sudo /opt/powerglove/bin/powerglove-pair
```

Leave that command running on RetroPie, then complete pairing at <https://UNO-Q-NAME.local:8443/setup>. The code is single use and expires after two minutes. Password pairing is also available when RetroPie accepts SSH password login; the password is used for one encrypted operation and is not stored.

The RetroPie token must contain at least 16 characters and should remain owned by `root`, readable by the `input` group, and inaccessible to other users:

```
SH
sudo chown root:input /etc/powerglove/token
sudo chmod 0640 /etc/powerglove/token
```

Recover without browser pairing

Use this fallback only when neither browser pairing method works. Both machines must already have the software installed.

- 1 In App Lab, open the active application's private `data/device.json` and locate its `token` value.
- 2 On RetroPie, run `sudo nano /etc/powerglove/token`. Replace the file contents with that same value on one line, without quotation marks. Do not enter it as a shell command.
- 3 Save with Ctrl+O, confirm the filename, and exit with Ctrl+X. Apply the ownership and permission commands above.
- 4 Run `sudo systemctl restart powerglove-receiver.service`, then test controller delivery from Dashboard. Clear the token from your clipboard and close the private file afterward.

If you generate a new token in Setup, pair the devices again immediately. Do not transfer the new token through a command-line argument; process listings and shell history can expose it.

RetroPie connection settings

The active launcher file is:

```
/etc/powerglove/launcher.json
```

It tells the runcommand hooks where to send profile changes when a game starts or exits.

```
JSON
{
  "uno_q": "UNO-Q-NAME.local",
  "port": 55356,
  "token_file": "/etc/powerglove/token",
  "registry": "/etc/powerglove/games.json",
  "timeout": 0.4
}
```

Field	Meaning
<code>uno_q</code>	UNO Q hostname or reserved address reachable from RetroPie.
<code>port</code>	RetroPie to UNO Q profile-control port. This is <code>55356</code> , not the controller-state port.
<code>token_file</code>	Protected shared-token file. Keep the token out of this JSON file.
<code>registry</code>	Active ROM-to-profile mapping.
<code>timeout</code>	Seconds to wait for each acknowledgement. The hook retries up to three times and never prevents a game from launching.

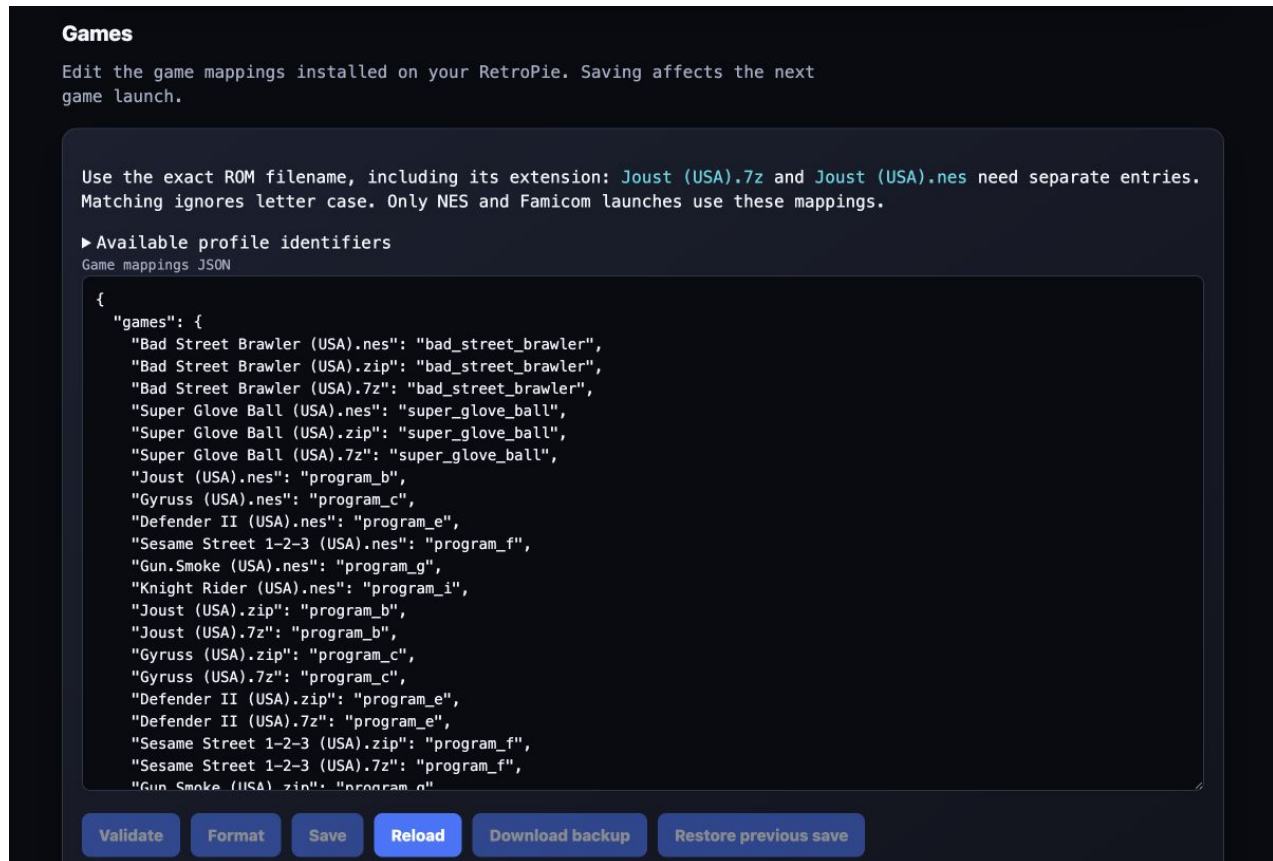
Validate changes before launching a game:

```
SH
python3 -m json.tool /etc/powerglove/launcher.json >/dev/null
```

Use a `.local` hostname when multicast DNS is reliable on your LAN. A reserved DHCP address is a useful fallback. Do not use an address that may later be assigned to another device.

Register games and select profiles

Edit mappings in Games



Games section below pairing on Setup

Games is a section of **Setup**, below pairing; it is not a separate navigation tab.

- 1 Open **Setup** -> **Games** in the UNO Q website. Both machines must be online and paired. The page reads the registry used by the installed RetroPie launch hook.
- 2 Select **Download backup** to keep a copy of the last verified installed registry on your computer.
- 3 Edit the JSON, adding the exact ROM filename and a supported profile identifier inside [games](#). Expand **Available profile identifiers** for the choices. Preserve your existing entries.
- 4 Select **Validate**. It checks JSON syntax, supported profiles, and duplicate filenames, including names that differ only by letter case. **Format** tidies the JSON without saving it.
- 5 Select **Save**. Wait for confirmation that RetroPie saved the file and the UNO Q read it back successfully.
- 6 Launch or restart the game and confirm its profile on Dashboard.

Saving does not change the current game's profile. **Restore previous save** swaps in the last valid version. **Reload** discards your draft after confirmation. If another editor changed the installed registry, saving is refused; download or copy

your draft before reloading. Connection failures leave the draft in the browser. Leaving or refreshing the page can discard unsaved work.

The Games section needs the [powerglove-games.service](#) installed by the current RetroPie setup workflow. If it reports an unavailable service, update the RetroPie installation, check pairing, and ensure TCP [55358](#) is reachable from the UNO Q. Games does not require an SSH password after pairing.

Registry format

The active game registry is:

```
/etc/powerglove/games.json
```

PowerGlove Vision matches the exact ROM basename, including its extension, without regard to letter case. Directory names are ignored. Automatic profile selection currently applies only when RetroPie reports the system as [nes](#) or [famicom](#); other systems turn gesture control off.

```
JSON
{
  "games": {
    "Bad Street Brawler (USA).nes": "bad_street_brawler",
    "Super Glove Ball (USA).zip": "super_glove_ball",
    "Joust (USA).nes": "program_b"
  }
}
```

Add the exact filename shown in EmulationStation. Zipped and 7-Zip copies need their own entries because [.nes](#), [.zip](#), and [.7z](#) are different basenames.

Included game	Profile
Bad Street Brawler	bad_street_brawler
Super Glove Ball	super_glove_ball
Joust	program_b
Gyruss	program_c
Defender II	program_e
Sesame Street 1-2-3	program_f
Gun Smoke	program_g
Knight Rider	program_i

The table above is the complete set of games recognized automatically by the shipped registry. Programs A, D, and H are fully implemented profiles rather than omitted games: [program_a](#) is a pinball control scheme, [program_d](#) reverses all four directions for challenge or accessibility use, and [program_h](#) provides general-purpose movement with pulsed buttons. They deliberately have no default ROM assignment.

Any appropriate NES or Famicom ROM can use one of those profiles after you add its exact basename to the [games](#) object. Every profile value must be one of the supported identifiers; an unknown value invalidates the registry.

```
SH
sudo python3 -m json.tool /etc/powerglove/games.json >/dev/null
```

Test profile communication independently of a game:

```
SH
sudo /opt/powerglove/bin/powerglove-profile \
  --uno-q UNO-Q-NAME.local \
  --token-file /etc/powerglove/token \
  --profile program_b
```

Use `--profile off` to stop gesture output. A successful request prints an acknowledgement and starts the matrix glove attract animation. In this healthy idle state the camera and MediaPipe tracker are closed, while the website and authenticated profile listener remain available.

Check a queued profile change

The launch helper reports **profile queued** when the UNO Q has authenticated and queued the request. Dashboard shows the applied profile and ROM name once the worker processes it. Camera startup can take longer than the acknowledgement; an acknowledgement does not mean that the camera is ready or that controller delivery is enabled.

App Lab's `local:profile_control` brick publishes UDP `55356` through a small relay to the worker. The relay holds no pairing token and forwards signed packets unchanged. Both this brick and `local:avahi_resolver` must be present in `app.yaml`, so the services return when App Lab regenerates its containers. The setup and Wi-Fi update helpers also add their includes to existing Compose configuration. On the UNO Q, check the published port with:

```
SH
docker port powerglove-vision-profile-relay-1 55356/udp
```

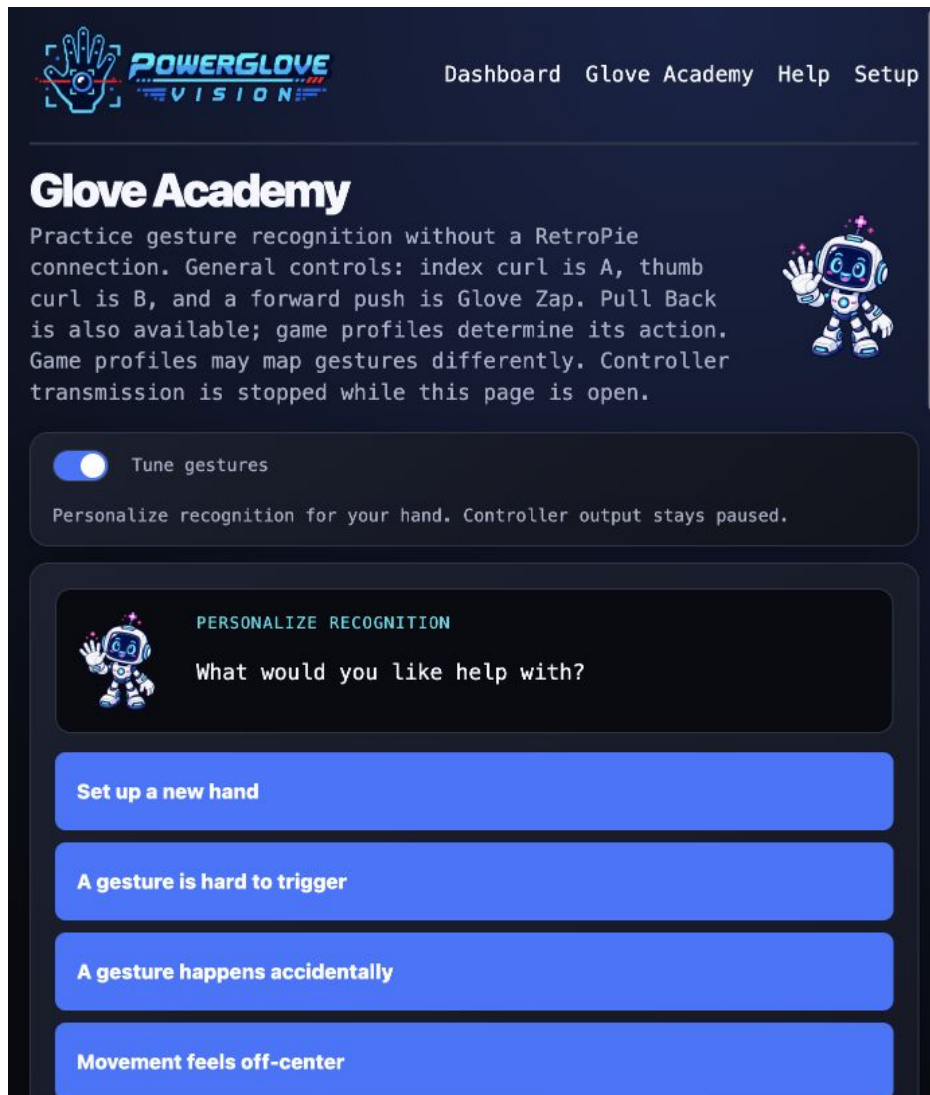
Expect a host binding for port `55356`. If it is missing, update the application and rerun UNO Q setup. On RetroPie, compare the game's actual filename, including `.nes`, `.zip`, or `.7z`, with `/etc/powerglove/games.json`. These are separate exact entries. Updating the template does not overwrite an installed registry; add missing names while preserving your custom mappings.

Tune gesture sensitivity

Use **Glove Academy -> Tune gestures** to personalize recognition. You do not need to edit `config/profiles.json`; it is the release-owned shared baseline. Updates back up and replace it. Personal adjustments belong in `data/gesture-tuning.json`, which remains untouched.

- 1 Choose **Set up a new hand**, **A gesture is hard to trigger**, **A gesture happens accidentally**, or **Movement feels off-center**.
- 2 Choose the gesture when asked. Off-center movement instead shows the saved center and an explicit **Set this as my center** action.
- 3 Keep the complete hand visible at 70% confidence for one second. Select **I'm ready** and wait through the two-second countdown.
- 4 Follow the three recordings. Ordinary poses and movement steps last two seconds. Glove Zap and Pull Back use a six-second middle step containing three motions and returns.

- 5 Analyze the recording and try the temporary preview twice. Return to neutral after each use and remain neutral for three seconds.
- 6 Save when the guided test passes. Only selected components are merged into the existing version-1 tuning file.



Tune mode with Pixel Pal guiding the personalization choices

The reference screenshot deliberately excludes the live camera area. Pixel Pal presents one instruction and primary action at a time. The numerical table and diagnostic capture are collapsed under **Advanced thresholds and diagnostics**. The matrix shows a scanning **T** while tuning and a matching scanning **L** in ordinary practice.

Activation is the point where a gesture begins; release is the lower point where it stops. Separate values prevent rapid on/off flickering. Gameplay movement mappings use these same held states, including wrist steering, push, pull-back, and braking; game-specific button assignments and pulses still apply. Directions and fingers can be adjusted independently. Compound gestures share component thresholds, so changing a finger also affects other gestures that use it. Suggested menu-pose adjustments tune the closed fingers; already extended fingers retain their existing settings from hand setup or existing personal/default values. Button assignments and menu hold timing remain unchanged.

Hand setup learns open and curled thresholds for all five fingers. Individual tuning can be used without setup; it only learns new thresholds for fingers observed both open and curled. Fingers extended throughout retain hand-setup thresholds or existing settings. Feedback uses the same V-sign and thumbs-up checks as recognition. Hand setup reset restores all five finger components; individual reset restores only the selected components. Difficult gestures place activation 55% into the

measured rest-to-action gap; accidental gestures use 75% activation and 40% release. Standard setup retains 65% activation and 30% release. Every path rejects a gap below 0.08.

Only adjusted components override all game profiles. Untuned components retain the shared supplied values. Personal adjustments are saved atomically in [data/gesture-tuning.json](#) and survive application restarts and normal updates. Normal personalization saves no images or recordings. Existing version-1 files remain compatible; hand setup adds ordinary finger pairs rather than a new file format. The versioned format is:

```
JSON
{
  "version": 1,
  "thresholds": {
    "index": {"on": 0.6, "off": 0.4}
  }
}
```

Each pair must contain finite numbers with $0 \leq \text{off} < \text{on}$. Finger and pull activation cannot exceed 1; wrist rotation cannot exceed 2; movement and push cannot exceed 4. These are normalized measurements, not distances in centimetres.

Tuning pauses controller delivery. A game launch may update the selected game but cannot interrupt tuning or send game input. Leaving Tune discards its preview; a disconnected browser's session expires after six seconds. Return to Dashboard and explicitly start controller delivery when ready to play. **Recalibrate neutral** changes the resting reference separately and invalidates any current recordings.

Private Academy diagnostic capture

The Advanced diagnostic is separate from personalization. Eight user-paced cues exercise neutral, directions, A/B, menu poses, rolls, depth motion, Menu Guard, and tracking recovery using the deployed **MediaPipe Hands (proven)** backend. The UNO Q records a temporary local AVI only while a cue is active. Completion produces an aggregate JSON report containing detection continuity, confidence, latency, hand brightness, and recognized state names. It contains no frames or per-frame landmarks. The AVI is deleted immediately after analysis or cancellation; an abandoned capture is deleted after 30 minutes. No network upload occurs.

Supplied shared recognition defaults

The following fields describe the shipped [config/profiles.json](#). They remain useful for understanding the defaults; personal tuning is managed through Glove Academy.

Threshold fields

Field	What it measures	Effect of lowering the value
<code>move_on</code>	Palm displacement from center, normalized by palm size	Movement activates sooner
<code>move_off</code>	Palm displacement at which active movement releases	Movement stays active farther back toward center
<code>coordinate_edge_margin</code>	Camera margin excluded from native X/Y travel	Native travel reaches its edge closer to the camera boundary
<code>coordinate_smoothing_min</code>	Minimum weight assigned to the newest native coordinate	Small native movements respond more immediately but may show more jitter
<code>coordinate_smoothing_max</code>	Maximum newest-coordinate weight during deliberate travel	Large native movements catch up less quickly
<code>coordinate_motion_boost</code>	How quickly movement raises the coordinate weight toward its maximum	Native travel receives less immediate acceleration
<code>curl_on</code>	Normalized finger curl, where <code>0</code> is straight and <code>1</code> is tightly curled	Curl actions activate with less bend
<code>curl_off</code>	Curl amount at which an active curl releases	Curl stays active until the finger is straighter
<code>roll_on</code>	Wrist rotation from the centred angle	Roll actions activate with less rotation
<code>roll_off</code>	Rotation at which active roll releases	Roll stays active closer to neutral
<code>push_on</code>	Relative increase in apparent hand size from center	Push actions activate with less forward movement
<code>push_off</code>	Depth change at which an active push releases	Push stays active closer to the centred depth
<code>depth_confirm_frames</code>	Consecutive beyond-threshold observations required for Glove Zap or Pull Back	Fewer observations accept shorter changes but reduce spike rejection

Field	What it measures	Effect of lowering the value
<code>depth_motion_window_ms</code>	Window in which the required depth travel must occur	A longer interval accepts slower depth motion
<code>depth_motion_delta</code>	Minimum normalized palm-scale travel toward or away from the camera	Smaller apparent-size changes can qualify as depth actions
<code>pulse_hz</code>	Repetition rate for profiles that pulse an action	Repeated actions become slower
<code>loss_release_ms</code>	Tracking-loss delay before all controls release	Controls release sooner after the hand disappears

For each gesture, keep the `_off` value lower than its `_on` value. The gap is hysteresis: it prevents a value near the activation point from rapidly turning on and off. A very large gap can make the control feel sticky.

The supplied shared recognition defaults are:

```
JSON
{
  "move_on": 0.28,
  "move_off": 0.14,
  "coordinate_edge_margin": 0.08,
  "coordinate_smoothing_min": 0.70,
  "coordinate_smoothing_max": 1.00,
  "coordinate_motion_boost": 4.00,
  "curl_on": 0.50,
  "curl_off": 0.35,
  "roll_on": 0.58,
  "roll_off": 0.40,
  "push_on": 0.34,
  "push_off": 0.18,
  "depth_confirm_frames": 2,
  "depth_motion_window_ms": 250,
  "depth_motion_delta": 0.10,
  "pulse_hz": 7.0,
  "loss_release_ms": 120
}
```

The `recognition` object applies to every game profile. During neutral calibration, the worker records ordinary X/Y jitter and raises only the relevant movement thresholds when necessary; it never lowers the `0.28` activation or `0.14` release baselines. Programs A-I and dedicated game profiles only decide how those shared recognition states map to controller output.

When adjusting numeric values in Tune, change one pair at a time in steps of approximately `0.02` to `0.05`, then test from the same camera position. Useful adjustments include:

- Recalibrate first if directional movement requires too much travel or moves at rest.
- Keep `move_off` below `move_on` so a direction releases promptly near center.
- Raise an `_on` value when an action triggers unintentionally.
- Increase `pulse_hz` when a repeating action is too slow.

- Keep `loss_release_ms` short enough to release safely but long enough to tolerate a few missed camera frames.

Saved personal tuning applies without reopening the camera and across every profile. The saved neutral calibration is reused; recalibrate only if your physical setup has changed or your resting hand position produces unwanted movement. Gesture-to-button assignments are implemented by each profile in the application; threshold changes adjust sensitivity but do not remap buttons.

RetroPie receiver and virtual controller

The receiver verifies authenticated UDP packets and creates a Linux `uinput` gamepad named **PowerGlove Vision**. Its installed service is:

```
/etc/systemd/system/powerglove-receiver.service
```

The supplied service listens on all local interfaces at UDP port `55355`, reads `/etc/powerglove/token`, and releases held controls when a socket receive times out after 250 milliseconds. This is a receive timeout, rather than a separate timer for the last valid packet. If you change the controller port in UNO Q Setup, add the same `--port` value to the service's `ExecStart`, then reload and restart:

```
SH
sudo systemctl daemon-reload
sudo systemctl restart powerglove-receiver.service
```

The companion `powerglove-receiver.timer` starts the receiver 45 seconds after boot. This lets EmulationStation finish its initial controller scan first. Keep the timer enabled and the service itself disabled for boot activation. Starting the receiver too early can cause frontend pauses and conflicts with other USB devices, such as a BitPixel display.

```
SH
systemctl is-enabled powerglove-receiver.timer
systemctl is-enabled powerglove-receiver.service
sudo systemctl status powerglove-receiver.service
sudo journalctl -u powerglove-receiver.service -n 100 --no-pager
```

The virtual gamepad appears only after the first authenticated controller packet. On the UNO Q, select **Start controller** and show a centred hand before deciding that the device is missing.

RetroArch autoconfiguration

Install the supplied mapping at:

```
/opt/retropie/configs/all/retroarch/autoconfig/PowerGlove Vision.cfg
```

It matches the virtual device name plus vendor and product IDs `1:1`, uses the `udev` input driver, and maps D-pad directions, A, B, Start, Select, and four axes to a standard RetroPad. It does not configure an I-PAC, 8BitDo controller, or any other physical controller.

If RetroArch has a hand-written override for this device, remove or reconcile that override before diagnosing the supplied autoconfiguration.

UNO Q privileged host helpers

The standard installation includes the host shutdown helper so Dashboard and Setup can halt Linux cleanly. It consists of two systemd units and one boot-time readiness rule:

Repository file	Installed path	Purpose
uno-q/powerglove-system-shutdown.path	/etc/systemd/system/powerglove-system-shutdown.path	Watches for one fixed shutdown request in the application's private data directory
uno-q/powerglove-system-shutdown.service	/etc/systemd/system/powerglove-system-shutdown.service	Removes that request and asks systemd to halt Linux cleanly
uno-q/powerglove-system-shutdown.conf	/etc/tmpfiles.d/powerglove-system-shutdown.conf	Recreates the unprivileged readiness marker at boot or after application replacement

The standard installer adds [powerglove-camera-recovery.path](#), its fixed-purpose service, [/usr/local/libexec/powerglove-camera-recovery](#), and a tmpfiles rule. Installation does not require a camera. If exactly one UVC camera is connected, the helper records it and its nearest external parent hub immediately; otherwise enrollment is deferred until vision first sees the camera successfully. The root-owned [/etc/powerglove-camera-recovery.json](#) allowlist stores the observed camera identity plus the hub identity and physical USB path. A later healthy sighting updates the association automatically if the camera has moved.

During an outage the unprivileged application can request only the helper's fixed operation. It validates the stored hub path and identity before resetting that hub, waits for one UVC camera to enumerate, updates the allowlist, and sets the camera and hub power policies to [on](#). It never guesses among hubs. A root-owned 60-second cooldown and the application's one-request-per-outage rule prevent reset loops. Resetting the hub can briefly interrupt USB Ethernet and any other devices attached to it. Before the first successful camera sighting, there is deliberately no reset target; reconnect or power-cycle the camera once to let automatic enrollment complete.

Install all host helpers with [scripts/install-uno-q-shutdown-helper.sh](#), or install/repair only camera recovery with [scripts/install-uno-q-camera-recovery-helper.sh](#). The standard installer also creates the private [data/.shutdown-enabled](#) marker that allows the web UI to offer the action and [data/.camera-recovery-enabled](#) for camera recovery. The tmpfiles rules restore both markers at boot. The application cannot use this mechanism to execute an arbitrary privileged command; it can only create the two fixed request files.

Do not change the request path in only one component. The web application, path unit, service, and marker must continue to agree. After installation, confirm that [powerglove-system-shutdown.path](#) and [powerglove-camera-recovery.path](#) are enabled and active.

Network ports and trust boundary

Keep PowerGlove Vision on a trusted home or cabinet LAN. Do not forward these ports through a router or expose them directly to the Internet.

Port	Direction	Purpose
UDP 55355	UNO Q to RetroPie	Authenticated live controller state
UDP 55356	RetroPie to UNO Q	Authenticated game-profile requests and acknowledgements
TCP 8088	Browser to UNO Q	Dashboard, Help, Glove Academy, and ordinary Setup UI, including Games
TCP 8443	Browser to UNO Q	TLS Setup and pairing workflow
TCP 55358	UNO Q to RetroPie	Paired game registry reads, saves, and restoration
TCP 55357	UNO Q to RetroPie	Temporary one-time-code pairing helper

The two UDP ports serve different purposes despite their similar numbers. The `port` in UNO Q `device.json` is normally 55355; the `port` in RetroPie `launcher.json` is normally 55356.

The ordinary dashboard is reachable by other devices on the LAN. Pairing and shutdown require additional protections, but the project assumes that the LAN itself is trusted. Review [SECURITY.md](#) before using a shared, guest, school, or public network.

UNO Q matrix artwork

The built-in display is a 13-by-8 monochrome blue LED matrix with eight brightness levels. Treat it as a very small dot-matrix display in the spirit of a pinball DMD or monochrome BitPixel display, rather than as a miniature screen. The sketch encodes status, pairing, profile identifiers, and the gestures-idle Power Glove attract sequence. Glove Academy and Tune share a dim letter, bright scan line, and trailing glow: eight frames at 160 milliseconds each (a 1.28-second cycle).

Matrix artwork should use broad motion, recognizable silhouettes, and strong separation between the subject and its brightest effect. A moving spark needs a dim body underneath it, a medium halo, and a full-bright core. Pulses should emphasize an outline instead of illuminating every pixel at maximum brightness, which erases the shape on the physical display. Transitional objects need to cross several columns and remain visible for more than one frame; isolated one-pixel changes are easily lost to persistence and viewing angle.

Always judge animation timing and grayscale on the physical UNO Q. A source grid or browser mock-up is useful for finding malformed frames, but it cannot reproduce LED bloom, exposure, or perceived persistence. A short video covering several complete loops is the preferred review artifact for later refinements.

Files most users should not edit

File or directory	Purpose
app.yaml	Arduino App Lab application metadata and exposed ports
sketch/sketch.yaml	UNO Q sketch platform and pinned Arduino library dependencies
pyproject.toml	Python package metadata, supported interpreter range, and optional dependencies
python/worker-wheels/	Platform-specific MediaPipe worker dependency supplied by the App Lab installation ZIP
data/models/hand_landmarker.task	Checksum-verified cached model, installed from the bundle when vision is first activated
data/uv-cache/ and data/uv-python/	Generated private worker runtime and package cache
.cache/app-compose.yaml	App Lab generated container configuration
data/.shutdown-enabled	Readiness marker installed by the fixed-purpose shutdown helper included in standard setup
data/.camera-recovery-enabled	Readiness marker for the fixed-purpose host USB-camera recovery helper
data/camera-recovery-request	Ephemeral one-shot request consumed by the root-owned camera recovery service
output/pdf/	Generated PDF editions; public editions are served by Help, while the cabinet quick reference remains private

Changing manifests can prevent App Lab from starting the application. Generated files and private runtime caches should not be committed or added to an App Lab installation ZIP. The installation ZIP is built as:

```
output/app-lab/PowerGlove-Vision-Uno-Q.zip
```

The installation ZIP includes the verified model at [models/hand_landmarker.task](#), its Apache 2.0 license, and third-party notices. It excludes private [data/](#), caches, tests, Git metadata, and the cabinet-specific quick-reference PDF. It includes only the allowlisted public PDF editions used by Help. When an active profile first needs vision, the application installs the bundled Google Hand Landmarker model into its private cache and verifies its SHA-256 checksum. A download is attempted only if the bundle is absent. The model stays unopened while **Gestures off** is selected.

Automated quality and package verification

Maintainers use [.github/workflows/quality.yml](#) for pull requests, pushes to [main](#) and [dev](#), and manual runs. The workflow tests the project on supported Python versions, checks Python and shell syntax, validates JSON, audits source and documentation, rebuilds and inspects the PDF set, and verifies the App Lab installation ZIP. The workflow has read-only repository permissions and uses no deployment credentials.

Ordinary installers do not need to edit this workflow. If you fork the project, keep live UNO Q credentials and cabinet deployment out of hosted CI; deployment should remain an explicit action from a trusted machine.

Back up and update safely

Back up custom configuration before replacing an installation:

Item	Why it matters
UNO Q data/calibration.json	Preserves your neutral hand position, size, and wrist angle; recalibrate if the physical setup changes
UNO Q data/device.json	Contains device settings and the private token
UNO Q config/profiles.json	Release-owned shared defaults; updates back up and replace this file
UNO Q data/gesture-tuning.json	Contains saved personal gesture sensitivity values and is preserved
RetroPie /etc/powerglove/games.json	Contains local ROM mappings
RetroPie /etc/powerglove/launcher.json	Contains local host and path settings
RetroPie /etc/powerglove/token	Contains the matching private token
RetroArch PowerGlove Vision.cfg	Contains any deliberate local mapping changes

Store token-bearing backups privately with restricted permissions. The supplied Wi-Fi deployment script preserves the UNO Q [data/](#) directory. The App Lab installation ZIP never contains your token or private model cache; it includes the unmodified public model.

After an update, confirm that the active files under [/etc/powerglove/](#) still contain your local hostnames and ROM names. Updating repository templates does not automatically migrate active configuration.

Troubleshooting by symptom

Symptom	Configuration checks
Dashboard works but no virtual controller appears	Start the controller, show a centred hand, verify the receiver service and shared token, then check UDP 55355 .
Controller appears but a game uses the wrong gestures	Confirm the system is nes or famicom and the exact ROM basename exists in /etc/powerglove/games.json .
Game launches slowly while UNO Q is offline	Confirm timeout remains near 0.4 ; the hook retries but must never block game launch indefinitely.
Profile command is not acknowledged	Check the UNO Q name, UDP 55356 , pairing token, and the UNO Q application status.
Gestures off shows a blinking X	Update PowerGlove Vision; Gestures off should show the glove attract animation and must not open the camera.
Camera disappears after reboot	Check lsusb and /dev/v4l/by-id/ , reconnect the camera or hub if absent, and keep Camera set to auto unless selecting a specific device. See startup diagnostics .
First activation is slow	Allow background preloading to finish and inspect the startup stage logs before attributing the delay to the camera.
Movement triggers too late	Recalibrate neutral first and verify the hand is steady; all profiles share the responsive movement thresholds.
Direction remains stuck	Recalibrate neutral, verify return toward center and tracking-loss release, then review the shared move_off value.
Pairing suddenly fails after a Setup change	A rotated token invalidates the old pairing; run the pairing flow again.
EmulationStation pauses or another USB device behaves unexpectedly at boot	Verify receiver startup is controlled by the 45-second timer and the service is not independently enabled at boot.

Configuration file catalog

Repository file	Active or installed copy	Used by
config/device.example.json	UNO Q application data/device.json	Vision supervisor and Setup UI
config/profiles.json	UNO Q application config/profiles.json	Gesture engine
config/games.json	RetroPie /etc/powerglove/games.json	Launch hook and profile selector
config/launcher.example.json	RetroPie /etc/powerglove/launcher.json	RetroPie launch and exit hooks
retropie/retroarch/PowerGlove Vision.cfg	RetroArch autoconfig directory	RetroArch input system
retropie/powerglove-receiver.service	/etc/systemd/system/	Privileged virtual-controller receiver
retropie/powerglove-games.service	/etc/systemd/system/	Paired Games editor service on RetroPie
data/gesture-tuning.json (runtime only)	UNO Q application data/gesture-tuning.json	Global personal threshold overlays
retropie/powerglove-receiver.timer	/etc/systemd/system/	Delayed boot activation
uno-q/powerglove-system-shutdown.path	/etc/systemd/system/	Fixed shutdown request watcher
uno-q/powerglove-system-shutdown.service	/etc/systemd/system/	Fixed clean-shutdown action
uno-q/powerglove-system-shutdown.conf	/etc/tmpfiles.d/	Boot-time shutdown readiness marker
uno-q/powerglove-camera-recovery.path	/etc/systemd/system/	Watches the fixed camera-recovery request
uno-q/powerglove-camera-recovery.service	/etc/systemd/system/	Runs the bounded camera recovery action
uno-q/powerglove-camera-recovery.py	/usr/local/libexec/powerglove-camera-recovery	Enrolls one UVC camera and resets only its last observed, identity-checked parent hub
uno-q/powerglove-camera-recovery.conf	/etc/tmpfiles.d/	Boot-time camera-recovery readiness marker

Repository file	Active or installed copy	Used by
Runtime camera allowlist	<code>/etc/powerglove-camera-recording.json</code>	Root-owned camera identity and last successfully observed hub path and identity
<code>.github/workflows/quality.yml</code>	GitHub Actions	Automated tests and release verification
<code>app.yaml</code>	UNO Q application root	App Lab
<code>sketch/sketch.yaml</code>	UNO Q application sketch directory	Arduino build system
<code>pyproject.toml</code>	Repository or deployed application root	Python packaging and tests

When a change goes wrong, restore the last known-good active file rather than copying every repository template over the installation. Validate JSON, restart only the affected service or application, and test packet delivery before changing controller mappings.

Local hostname resolution inside App Lab

If the console name fails, use **Test console name** in Connection, then follow [hostname troubleshooting](#). A router-reserved IPv4 address is a fallback, not a setup requirement.

Both machine installers install `avahi-daemon` and `libnss-mdns` and enable Avahi at boot. The UNO Q host dependency supports native hostname lookups; it does not replace the app-owned resolver used inside the container. Setup check mode verifies the dependency and Avahi service, and the UNO Q check tests the configured destination inside the app.

The app-owned `local:avahi_resolver` brick survives App Lab container regeneration. It mounts `/run/avahi-daemon` read-only in a separate unprivileged service and exposes only IPv4 `.local` queries through `data/.avahi-resolver.sock`. It has no network interface or published port. The app prefers this private socket; the direct host socket remains a compatibility fallback. Both sockets are runtime files, not configuration to back up or distribute.

All gameplay and pairing lookups use this resolver. Answers expire after five seconds, so DHCP changes do not require editing an address. Ordinary DNS names use the system resolver. Generic container `getent` is not the app's mDNS test; use Connection's hostname test or the setup command's check mode.

Known limitation: UNO Q restarts after Shutdown

On the tested UNO Q, a graceful `halt` still leads to an automatic restart, including when connected directly to a Mac without the powered hub. The helper requests halt correctly, but remaining stopped is not verified. Do not use loss of the website or a fixed delay as confirmation that power can safely be removed. See the installation guide for the investigation status and Arduino guidance.

Saved neutral-hand calibration

The worker saves its completed neutral reference in `data/calibration.json`. It includes palm position, apparent size, wrist angle, and normal X/Y positional jitter; it is not a personally trained gesture model. The jitter estimate can raise the shared movement thresholds above their baseline, but never makes them more sensitive. Glove Academy, gameplay, profile changes, camera reconnects, and worker restarts reuse this reference. **Calibrate** explicitly replaces it after sampling completes; an interrupted calibration preserves the previous saved reference. Recalibrate after moving your camera or changing your seating or standing position.

On first use, or if the saved file is missing or invalid, the worker samples an initial reference automatically. Calibration requires 24 complete observations at 70% hand confidence or better. It averages palm center and apparent size, uses a circular mean for wrist angle, and records the 95th-percentile X/Y deviation as normal jitter. Hold a relaxed open hand still at the intended neutral point and distance. Repeating from the same physical setup should produce a close reference, not identical floating-point values, because camera landmarks vary from frame to frame.

The release-owned `config/profiles.json` contains the portable starting point: movement thresholds, coordinate range, stabilization, finger, roll, depth, pulse, and tracking-loss defaults. The installer's package validator requires that file and the recognition modules, and updates back up and replace the shared profile file. It deliberately excludes `data/calibration.json`; copying one person's palm coordinates and apparent scale to another camera would create a misleading neutral point. Existing neutral calibration and personal gesture tuning remain preserved during updates. A storage failure is reported as `calibration_save_error` in status; the reference remains usable in memory but will not survive a worker restart.

Command-line reference

Use this section to look up a flag without interrupting the installation steps. It covers every project command and script, plus the external-command options used in these guides. Defaults describe this source version, not a guarantee about a future release. For external tools' other options, use their own help or manual pages.

An **option** begins with `-` or `--`. A **positional argument** is a value supplied in a particular place, such as `retropie` after `setup-machine.py`. Replace example hostnames and file paths with your own. Square brackets in usage descriptions mean optional; do not type them. Options that take a value need both the option and its value, such as `--port 55355`.

Install or inspect a machine

Run `sudo python3 scripts/setup-machine.py MACHINE [OPTIONS]` from the project directory on the target Linux machine. This is the recommended installer for both UNO Q and RetroPie. `--check` performs read-only checks; using `sudo` also lets those checks read protected token files.

Argument or flag	Default	Meaning
<code>MACHINE</code>	Required	<code>retropie</code> or <code>uno-q</code> ; selects the machine to install or inspect.
<code>--peer HOST</code>	None	Required for a new RetroPie launcher configuration; supplies the UNO Q hostname or IPv4 address. Existing launcher settings are preserved. On UNO Q, it prints guidance but does not change the saved receiver address.
<code>--check</code>	Off	Checks the existing installation without installing, restarting, or changing it.
<code>-h</code> , <code>--help</code>	-	Prints usage and exits.

Exit codes are **0** for success, **1** for an installation/check failure, and **2** for outstanding user action. Argument errors also use argparse's exit code **2**. The current check always asks for human gameplay confirmation.

Run the RetroPie receiver

Use `/opt/powerglove/bin/powerglove-receiver` on RetroPie. Normal operation is managed by its systemd service. A manual receiver cannot share the same port with that service: stop the service before a manual diagnostic run, then restart it afterward. Use `--token-file` rather than placing a token in shell history.

Flag	Default	Meaning
<code>--listen ADDRESS</code>	<code>127.0.0.1</code>	Local address to bind. <code>0.0.0.0</code> accepts packets on all local IPv4 interfaces, as the installed service requires.
<code>--port NUMBER</code>	<code>55355</code>	UDP port for controller packets; must match UNO Q settings.
<code>--token VALUE</code>	None	Supplies the shared token directly. Use only as an advanced alternative; the value can appear in process arguments.
<code>--token-file PATH</code>	None	Reads the shared token from a protected file. Supply exactly one of this flag and <code>--token</code> . The token must contain at least 16 characters.
<code>--timeout-ms NUMBER</code>	<code>250</code>	Socket receive timeout in milliseconds; a timeout releases held controls. Use a positive value.
<code>--native-state PATH</code>	<code>/run/powerglove/native-state</code>	Versioned latest-sample record for the optional custom Nestopia research core. Failure to create it does not disable FCEUmm/uinput.
<code>--dry-run</code>	Off	Prints received controls instead of creating a virtual input device.
<code>-h, --help</code>	-	Prints usage and exits.

Example: inspect packets without sending input to Linux. Run each command on RetroPie and press Ctrl+C to end the diagnostic receiver before restarting the service.

```
SH
sudo systemctl stop powerglove-receiver.service
sudo /opt/powerglove/bin/powerglove-receiver --listen 0.0.0.0 --token-file /etc/powerglove/token --dry-run
sudo systemctl start powerglove-receiver.service
```

Start one-time-code pairing

Use `/opt/powerglove/bin/powerglove-pair` on RetroPie with `sudo`. It opens a temporary TLS server, prints a code, installs the received token, and restarts the receiver. Complete the browser steps while it is running.

Flag	Default	Meaning
<code>--listen ADDRESS</code>	<code>0.0.0.0</code>	Local IPv4 address for the temporary server.
<code>--port NUMBER</code>	<code>55357</code>	Pairing server TCP port. The browser pairing client uses the standard port; keep the default for that workflow.
<code>--token-file PATH</code>	<code>/etc/powerglove/token</code>	Destination for the paired token; keep it aligned with the receiver's token file.

Flag	Default	Meaning
<code>--timeout SECONDS</code>	120	Lifetime of the pairing server. Use a positive value; the code is single use and attempts are limited.
<code>--receiver-service NAME</code>	<code>powerglove-receiver.service</code>	systemd service to restart after pairing succeeds.
<code>-h, --help</code>	-	Prints usage and exits.

The command returns `0` after pairing completes or `2` when the pairing window times out. Other failures are reported as errors.

Select a profile from RetroPie

Use `/opt/powerglove/bin/powerglove-profile` to test profile selection. It sends an authenticated request to the UNO Q. This changes the active profile; it does not change the saved startup profile or turn controller delivery on.

Flag	Default	Meaning
<code>--uno-q HOST</code>	Required	UNO Q hostname or IPv4 address.
<code>--port NUMBER</code>	55356	UNO Q profile-control UDP port.
<code>--token VALUE</code>	None	Supplies the token directly; prefer the protected-file option.
<code>--token-file PATH</code>	None	Reads the token from a file. Supply exactly one token option.
<code>--registry PATH</code>	<code>/etc/powerglove/games.json</code>	Registry used when <code>--profile</code> is omitted.
<code>--system NAME</code>	<code>nes</code>	System used for registry selection and request metadata. Automatic selection accepts <code>nes</code> or <code>famicom</code> .
<code>--rom PATH</code>	Manual selection	ROM path or filename used for registry lookup and displayed game metadata.
<code>--profile NAME</code>	None	Overrides registry lookup. Accepts <code>program_a</code> through <code>program_i</code> , <code>bad_street_brawler</code> , <code>super_glove_ball</code> , or <code>off</code> .
<code>--timeout SECONDS</code>	0.4	Wait for each acknowledgement; the sender makes up to three attempts.
<code>-h, --help</code>	-	Prints usage and exits.

Exit codes are `0` for acceptance, `2` for a timeout, and `3` for rejection. Acceptance means the request is queued; confirm the applied profile on Dashboard. Malformed arguments and file errors can also stop the command.

Run the paired Games service

The RetroPie installer starts this service automatically. Its installed command is `powerglove-games`; developers can also run `python3 -m powerglove_vision.game_registry`.

Flag	Default	Purpose
<code>--settings</code>	<code>/etc/powerglove/launcher.json</code>	Read the administrator-configured registry and token-file locations.
<code>--listen</code>	<code>0.0.0.0</code>	Bind address for the paired LAN service.
<code>--port</code>	<code>55358</code>	TCP port. The UNO browser proxy uses the standard port.
<code>--help</code>	Not applicable	Print usage without starting the service.

The service uses a single request handler with a three-second socket timeout, 64-KiB registry limit, and 64 pending challenges. Challenges expire after fifteen seconds and can be used once. The previous registry is saved beside the active file as `games.json.previous` (or the configured filename plus `.previous`). Installer-generated systemd permissions allow writing to the configured registry directory. Rerun installation if an administrator changes that directory.

Forward game-launch events

The installer connects `powerglove-retropie-hook` to RetroPie automatically. Use its wrapper scripts from existing runcommand hooks; do not run them again as an extra installation step.

Argument or flag	Default	Meaning
<code>ACTION</code>	Required	<code>start</code> selects a registered game's profile; <code>end</code> turns gestures off.
<code>SYSTEM</code>	Empty	Optional first metadata argument, such as <code>nes</code> .
<code>EMULATOR</code>	Empty	Optional second metadata argument supplied by RetroPie; accepted for compatibility.
<code>ROM</code>	Empty	Optional third metadata argument; its filename selects the profile.
<code>COMMAND</code>	Empty	Optional fourth metadata argument; accepted for compatibility. Quote it as one argument.
<code>--settings PATH</code>	<code>/etc/powerglove/launcher.json</code>	Reads destination, token path, registry path, and timeout settings.
<code>-h, --help</code>	-	Prints usage and exits.

`retropie/runcommand-onstart-powerglove.sh` forwards RetroPie's four positional arguments.

`retropie/runcommand-onend-powerglove.sh` needs none. Neither wrapper has its own flags. The executables in `retropie/bin/` forward their arguments to the corresponding Python commands.

Run the standalone vision tracker

`powerglove-vision` is the advanced camera worker, not the full App Lab website. Install the package's `vision` dependencies in a compatible Python environment before using it. Normal UNO Q use should start through App Lab instead.

Flag	Default	Meaning
<code>--receiver HOST</code>	Required	RetroPie hostname or IPv4 address. An empty string permits local-only tracking but cannot deliver controls.

Flag	Default	Meaning
<code>--port</code> NUMBER	55355	Destination UDP controller port.
<code>--token</code> VALUE	Required	Shared token. This worker currently has no <code>--token-file</code> flag; its token appears in process arguments.
<code>--profile</code> NAME	<code>bad_street_brawler</code>	Initial profile, one of the eleven supported profiles or <code>off</code> .
<code>--camera</code> VALUE	<code>auto</code>	Camera selection; use <code>auto</code> or a camera index.
<code>--width</code> PIXELS	640	Requested capture width; the camera may negotiate another size.
<code>--height</code> PIXELS	480	Requested capture height.
<code>--fps</code> NUMBER	60	Requested capture rate; not a guarantee of tracking or game frame rate.
<code>--camera-format</code> VALUE	MJPEG	Requested V4L2 format, either <code>MJPEG</code> or <code>YUYV</code> . Keep <code>MJPEG</code> for normal use; compare both only with the performance readings on hardware that advertises them.
<code>--inference-threads</code> NUMBER	4	CPU threads requested for each legacy MediaPipe inference calculator. Benchmark before changing.
<code>--tracker-backend</code> VALUE	<code>legacy</code>	<code>legacy</code> selects MediaPipe Hands (proven) ; <code>tasks-video</code> selects MediaPipe Tasks Video (experimental) using the packaged Hand Landmarker model. The identifiers remain stable for scripts.
<code>--preview-fps</code> NUMBER	5	Maximum rate at which optional browser preview jobs are submitted.
<code>--glove-color</code> VALUE	<code>none</code>	<code>none</code> , <code>white</code> , or <code>black</code> ; an informational label, not a different recognition model.
<code>--no-mirror</code>	Off	Disables horizontal image mirroring.
<code>--config</code> PATH	Project <code>config/profiles.json</code> , if present	Alternative gesture-threshold file. Otherwise built-in defaults are used.
<code>--model</code> PATH	Default verified model	Alternative MediaPipe Tasks model used only with <code>--tracker-backend tasks-video</code> . The standard <code>hand_landmarker.task</code> filename uses the verified model workflow.
<code>--web-host</code> ADDRESS	0.0.0.0	Address for the worker's diagnostic web server.
<code>--web-port</code> NUMBER	8088	Worker diagnostic port. App Lab overrides this to <code>8089</code> on loopback behind its main web server.
<code>--no-matrix</code>	Off	Disables direct matrix integration. App Lab uses this because its supervisor controls the matrix.
<code>--profile-listen</code> ADDRESS	0.0.0.0	Local IPv4 address for authenticated profile requests.
<code>--profile-port</code> NUMBER	55356	UDP port for profile requests.
<code>--controller-enabled</code>	Off	Starts sending controller input immediately; omit it for local inspection.
<code>-h</code> , <code>--help</code>	-	Prints usage and exits.

Deploy or repair the UNO Q application

Run these scripts from the project checkout on your development computer. Always supply your own SSH destination; the built-in fallback names the maintainer's board and is not appropriate for other installations.

Script or setting	Arguments and defaults	Effect
<code>scripts/deploy-uno-q-wifi.sh</code>	Optional positional <code>USER@HOST</code> ; <code>-h</code> or <code>--help</code> ; optional <code>UNO_Q_SSH_IDENTITY</code> private-key path	Transfers the Linux application, preserves <code>data/</code> , restarts the container, and checks web routes. Does not update RetroPie or flash the matrix sketch.
<code>UNO_Q_SSH_TARGET</code>	Environment variable; overridden by a positional destination	Sets the SSH destination. Without either setting, deployment falls back to the maintainer's board.
<code>UNO_Q_APP_DIR</code>	Environment variable; default <code>/home/arduino/ArduinoApps/powerglove-vision</code>	Remote deployment directory. Changing it does not change the shutdown helper's fixed path or the machine installer's path requirement.
<code>scripts/install-uno-q-shutdown-helper.sh</code>	Optional positional <code>USER@HOST</code> ; <code>-h</code> or <code>--help</code>	Installs the fixed shutdown watcher, service, and readiness rule. Uses the positional destination, then <code>UNO_Q_SSH_TARGET</code> , then the maintainer's fallback. The application directory is fixed.

Both scripts accept at most one destination. Deployment needs existing SSH key access; helper installation can prompt for a password through the terminal.

Build, inspect, or maintain project files

These scripts run from a full development checkout. Scripts marked **no flags** do not implement `--help`; adding it is not a safe way to inspect them because they may still perform their normal work.

Script	Arguments and flags	Result or requirement
<code>scripts/build-app-lab-package.sh</code>	No flags or positional arguments	Builds <code>output/app-lab/PowerGlove-Vision-Uno-Q.zip</code> ; requires Bash, rsync, zip, and the existing public PDFs.
<code>scripts/verify-app-lab-package.py</code>	Optional <code>ARCHIVE</code> path; <code>-h</code> , <code>--help</code>	Checks the supplied ZIP or the default ZIP above; prints its SHA-256. Returns <code>0</code> on success, <code>1</code> on verification failure.
<code>scripts/check-documentation.py</code>	<code>--require-pdfs</code> ; <code>-h</code> , <code>--help</code>	Checks Markdown, links, and coverage. The optional flag also inspects the PDF set and needs <code>pypdf</code> . Returns <code>0</code> on success, <code>1</code> on failure.
<code>scripts/check-source-docs.py</code>	No flags or positional arguments	Checks source headers and docstrings; returns <code>0</code> on success or <code>1</code> on failure.
<code>scripts/build-docs-pdf.py</code>	No flags or positional arguments	Rebuilds all registered PDF editions; requires ReportLab. Use only when ready to regenerate the PDFs.

Script	Arguments and flags	Result or requirement
scripts/build-nestopia-power-glove.sh	Optional build-directory positional argument	Clones a pinned official Nestopia revision, applies the isolated research patch, and builds a separately named core. It does not install or promote the core.
scripts/build-fceumm-benchmark.sh	Optional build-directory positional argument	Builds a pinned stock FCEUmm core in an isolated directory for the direction-response comparison. It does not install the core.
scripts/install-nestopia-powerglove.sh	Optional build-directory positional argument	Run with sudo on RetroPie after exact-ROM validation. Builds and installs only lr-nestopia-powerglove , plus its upstream GPLv2 license and distribution note; stock Nestopia remains untouched. The normal RetroPie installer offers this step when a registered Super Glove Ball ROM is found.
scripts/configure-super-glove-ball-core.py	<code>--rom PATH</code> <code>--mode MODE</code> <code>[--apply]</code> , where MODE is native or fceumm	Previews or atomically selects the custom core for one Super Glove Ball ROM. <code>--mode fceumm</code> is the explicit rollback.
scripts/run-nestopia-powerglove-trace.py	Core, exact ROM, trace/state/scratch paths	Runs controlled native phases, records the ROM digest and packet evidence, and can save temporary validation frames.
scripts/benchmark-direction-response.py	Paths to both cores and the exact Super Glove Ball ROM, scratch path, optional FCEUmm reference ROM, frame count, and JSON output	Runs matched-savestate activation and release comparisons for the same ROM in native and FCEUmm modes. The optional reference lane uses Gun.Smoke. ROMs and scratch output remain outside the project.
scripts/record-vision-benchmark.py	Optional camera, output, size, and frame-rate flags	Records a fixed 30-second, local-only cue sequence for near/far recognition, X/Y travel, jitter, depth, and recovery comparisons. It is never run by installation or used for training.
scripts/guided-vision-benchmark.py	Optional camera, output, bind address, port, size, and frame-rate flags	Serves a temporary live-preview page for user-paced, per-step benchmark recording. Each selected step has a two-second countdown; pauses between steps are not recorded. The camera is released when capture completes. Output stays local and is not training data.
scripts/benchmark-vision-replay.py	Local clip, required JSON output, and optional Tasks model path	Replays the same full frames through MediaPipe Hands at 1, 2, and 4 threads and through optional Tasks Video, at 640×480 and full-field 512×384, with preview closed and open. Reports p50/p95 inference, continuity, cue recognition, neutral false activations, coordinate jitter, and preview cost.
scripts/build-gesture-crops.py	No flags or positional arguments	Regenerates action illustrations from the gesture sheets; requires Pillow.
scripts/fetch-runtime-assets.sh	No flags or positional arguments	Installs and verifies the bundled model into project data/models/ ; downloads only if absent. Requires Python 3, plus curl for fallback downloads.

Script	Arguments and flags	Result or requirement
<code>scripts/configure-uno-q-mdns.py</code>	Required positional path to the generated Compose file; no flags	Internal installer/deployment helper that edits that file. Prefer the supported setup command.
<code>scripts/profile-relay.py</code>	No flags or positional arguments	Internal unprivileged UDP relay; publishes port 55356 and forwards to the worker, with bounded packet size, outstanding requests, and timeouts.
<code>scripts/avahi-resolver-service.py</code>	No flags or positional arguments	Internal service started by the resolver brick; opens its fixed Unix socket.
<code>python/main.py</code>	No flags or positional arguments	App Lab entry point; reads <code>data/device.json</code> and supervises the worker.
<code>python/ssh_pair.py</code>	No command-line flags	Internal password-pairing helper; receives its request through standard input. Use secure Setup instead.

Understand the system commands in these guides

The table explains the subcommands and flags shown in the documentation. Defaults and additional flags for these external programs depend on their installed versions; use `man TOOL` or that tool's `--help` for its full manual.

Command or syntax	Meaning of the options used here
<code>git clone --branch dev URL</code>	Downloads a repository and checks out <code>dev</code> . A final directory argument chooses the destination folder.
<code>git branch --show-current</code>	Prints the checked-out branch name.
<code>git status --short</code>	Shows a compact list of local changes.
<code>git pull --ff-only</code>	Updates only if Git can fast-forward; refuses an automatic merge when histories have diverged.
<code>cd PATH, cd ~, pwd</code>	Changes directory, opens your home directory, or prints the current directory.
<code>command -v NAME</code>	Prints where the shell finds a command; no path means it is unavailable.
<code>sudo COMMAND</code>	Runs a command with administrator privileges. Enter your account password only at the terminal prompt.
<code>apt update; apt install -y PACKAGES</code>	Refreshes package information, then installs packages. <code>-y</code> accepts package-manager confirmation prompts.
<code>python3 -m MODULE</code>	Runs a Python module. Used with <code>venv</code> , <code>pip</code> , <code>unittest</code> , <code>json.tool</code> , and <code>compileall</code> .
<code>python3 -m venv DIRECTORY</code>	Creates an isolated Python environment at that directory.
<code>. .venv/bin/activate</code>	Activates that environment in the current shell. The initial dot is a shell command.
<code>pip install -e PATH</code>	Installs a source checkout in editable mode. <code>[vision]</code> and <code>[receiver]</code> request optional dependency groups.

Command or syntax	Meaning of the options used here
<code>pip install -U PACKAGE;</code> <code>--upgrade</code>	Upgrades the named package; the two options have the same meaning.
<code>unittest discover -s tests -v</code>	Finds tests in <code>tests</code> (<code>-s</code>) and prints each test's result (<code>-v</code>).
<code>compileall -q DIRECTORIES</code>	Checks/compiles Python files; <code>-q</code> reduces routine output.
<code>json.tool FILE</code>	Parses and formats JSON; a parse error means the example or configuration is invalid.
<code>PYTHONPATH=src COMMAND</code>	Sets the import path for that command only, allowing it to use the local <code>src/</code> tree.
<code>ssh-keygen -t ed25519</code>	Creates an SSH key pair using the Ed25519 algorithm (<code>-t</code>). Follow the prompts without overwriting an existing key.
<code>cat FILE</code>	Prints a text file. In key setup, print only the <code>.pub</code> public key.
<code>ssh USER@HOST</code>	Opens a remote terminal. <code>exit</code> closes that session.
<code>ssh -o BatchMode=yes</code> <code>USER@HOST COMMAND</code>	Runs a remote command without interactive password prompts; useful for checking key access. <code>-o</code> supplies an SSH setting.
<code>ssh -t USER@HOST COMMAND</code>	Allocates a remote terminal, allowing an interactive remote prompt.
<code>docker port CONTAINER</code> <code>55356/udp</code>	Shows the host binding for the container's UDP profile port. This command does not change the container.
<code>systemctl status NAME</code>	Displays a service or timer's state. Press <code>q</code> if it opens a pager.
<code>systemctl start, stop, restart</code> <code>NAME</code>	Starts, stops, or restarts the named unit now.
<code>systemctl enable, disable NAME</code>	Enables or disables activation at boot. <code>--now</code> also starts or stops it immediately.
<code>systemctl is-enabled,</code> <code>is-active NAME</code>	Checks boot enablement or current activity.
<code>systemctl daemon-reload</code>	Reloads unit definitions after service files change.
<code>systemctl --no-block halt</code>	Requests a halt and returns without waiting. This is the helper's action, not a health check.
<code>journalctl -u NAME -n 100</code> <code>--no-pager</code>	Shows logs for a unit (<code>-u</code>), limits output to the last 100 entries (<code>-n</code>), and avoids a pager.
<code>systemd-tmpfiles --create</code> <code>FILE</code>	Applies creation rules from the named file, such as the readiness-marker rule.
<code>grep -A8 -B2 PATTERN FILE</code>	Shows matching lines with eight following lines (<code>-A</code>) and two preceding lines (<code>-B</code>).
<code>curl -sS URL</code>	Fetches a URL quietly (<code>-s</code>) while still showing errors (<code>-S</code>).
<code>ls -l PATH</code>	Lists files with permissions and ownership. A trailing directory slash refers to its contents.
<code>install -d -m MODE PATH</code>	Creates a directory (<code>-d</code>) with the specified permissions (<code>-m</code>). Without <code>-d</code> , <code>install</code> copies a file; <code>-o</code> sets its owner and <code>-g</code> its group.
<code>chmod MODE PATH; chown</code> <code>USER:GROUP PATH</code>	Changes permissions or ownership. Modes used here: <code>0600</code> owner only, <code>0640</code> owner read/write and group read, <code>0644</code> public read with owner write, <code>0700</code> private directory, <code>0755</code> publicly traversable directory or executable.
<code>rm FILE; rm -f FILE</code>	Deletes a file; <code>-f</code> suppresses prompts and ignores a missing file. Check the path before use.

Command or syntax	Meaning of the options used here
<code>nano FILE</code>	Opens a text editor. Save with Ctrl+O, confirm the name, and exit with Ctrl+X.
<code>modprobe uinput</code>	Loads the Linux virtual-input kernel module.
<code>tee FILE</code>	Writes its input to a file as well as the terminal. Used with <code>sudo</code> for a protected destination.
<code>printf '%s\n' VALUE</code>	Prints a value followed by a newline.
<code>ping HOST</code>	Tests whether a host replies to network echo requests; press Ctrl+C to stop. It does not test the application.
<code>hostname; hostname -I</code>	Prints the host's name, or its interface addresses with <code>-I</code> .
<code>bash -n FILE; sh -n FILE</code>	Checks shell syntax without running the file.
<code>> /dev/null</code>	Discards normal output while leaving errors visible.
Pipe operator	Sends the first command's output into the second command. The vertical bar is shell syntax.
<code>\</code> at the end of a command line	Continues the same command on the next line.
<code>"\$1" through "\$4"</code>	Quoted positional arguments forwarded by RetroPie's wrapper scripts.
<code>for ...; do ...; done</code>	Repeats the enclosed commands for each value.
<code>[-r FILE]</code>	Tests whether a file is readable. <code>continue</code> skips the current loop item.
<code>\${device##*/}</code>	Extracts the last component of a path in the camera-inspection loop.

Commands shown inside a source script may use additional internal options. Those are implementation details, not extra arguments accepted by the wrapper.

Tuning measurements and interface contract

Each three-second step needs at least twelve accepted measurements. Sampling requires a calibrated, detected hand with confidence at least 0.7; duplicate frames and non-finite measurements are excluded. Missing or low-confidence tracking does not contribute samples. Insufficient samples require retrying the step. Changing neutral calibration clears recordings and preview values.

Suggestions use the 95th percentile of the two open/rest recordings and the 10th percentile of the performed recording. The gap must be at least 0.08; activation is placed 65% into that gap and release 30% into it. An overlap error identifies the affected component, and no suggestion is retained. Fully return to the starting pose and distance before recording the final step.

For V-sign and thumbs-up, extended-only fingers retain existing thresholds; these samples cannot establish their curled boundary. Personal activation values check curled fingers and personal release values check extended fingers. Without personal settings, the legacy menu cutoffs in the table above apply. Live feedback and actual pose recognition use the same checks, with the existing deliberate hold retained. Before accepting an automatic suggestion, the app applies the candidate thresholds to all required fingers in the recordings. At least 90% of accepted samples must match the complete gesture together; the first and final recordings must likewise show all selected fingers extended. This allows limited tracking noise without letting a strong curl compensate for a bent extended finger. A failed check identifies the finger and phase and clears the preview.

V-sign requires extended index/middle and curled ring/pinky; thumbs-up requires an extended thumb and four curled fingers. Thumb extension measures straightness, not direction toward the top of the image. Individual tuning keeps

extended-only thresholds; use Set up my hand if comfortable extension needs adjustment. Preview and live gameplay testing still matter, and manual numerical edits remain a separate path rather than an automatically verified recording.

The status response's `tuning` object retains its version-1 fields and adds `problem`, `wizard_step`, `recipe`, `stable_ready`, `test`, `image_quality`, and `diagnostic`. `gesture`, `total_phases` (3), `completed_phases`, and `finger_feedback` continue to describe the active recipe. Each finger entry has `expected` (`extended` or `curled`), `matches`, `value`, and `threshold`. `saved`, `effective`, and `preview` distinguish persisted values, currently used pairs, and temporary adjustments. The six-second browser lease expires temporary recordings and previews; it does not delete saved thresholds. Manual preview and selective reset remain available under Advanced; normal wizard saving requires the guided activation, release, and neutral check.

Verified sketch dependencies

`sketch/sketch.yaml` pins Arduino Zephyr **1.0.0** for `arduino:zephyr:unoq`, Arduino_RouterBridge **0.4.3**, Arduino_RPClite **0.3.0**, ArxContainer **0.7.0**, ArxTypeTraits **0.3.2**, DebugLog **0.8.4**, and MsgPack **0.4.2**. Keep this complete configuration synchronized with the installed app. Platform installation, compile-only validation, and firmware upload are separate operations; see [Build and install matrix firmware](#).

Bad Street Brawler Glove Zap

The dedicated `bad_street_brawler` profile translates each forward-push activation into simultaneous Left + Right for 180 milliseconds. It suppresses other D-pad directions and A/B during that pulse. Holding the push does not repeat it; cross the saved push Release threshold before activating again. Menu poses, tracking loss, and recalibration cancel the pulse. Other game profiles are unchanged. The game, rather than the gesture engine, enforces its once-per-round allowance.

FCEUmm normally removes opposing directions. For this game, set `fceumm_up_down_allowed = "enabled"` in its game-specific options file. On the cabinet this is `/opt/retropie/configs/all/retroarch/config/FCEUmm/Bad Street Brawler (USA).opt`. The filename omits the ROM/archive extension. Preserve any existing game options; when creating the file, start from the current FCEUmm options to avoid changing unrelated sound, video, or input settings. Leave the global value unchanged.

In RetroArch, the equivalent workflow is to load the game, open Quick Menu's Core Options, allow opposing directions, and save a Game Options file. Verify that the game-specific file is loaded on the next launch. The existing receiver and PowerGlove gamepad mapping already carry separate Left and Right buttons; no R2/extra-trigger binding is needed. Disable the option for this game or remove its new options file to undo that part of the change.

The secret Left + Right attack is documented by the runner in [Speed Demos Archive](#). [FCEUmm's input code](#) shows the opposing-direction filter. Headless startup can verify options loading; confirm the attack, normal movement, and once-per-round behaviour in live play.

Repeatable RetroPie check and setup

The installer copies `scripts/configure-bsb-zap.py` and installs the `powerglove-bsb-zap` command. For an existing installation, copy the helper into `/opt/powerglove-src/scripts/` and its matching wrapper from `retropie/bin/` into `/opt/powerglove/bin/`. This is a separate, explicit game setup step; the main installer does not change game options automatically.

Run on RetroPie, using the exact ROM/archive path installed on your system:

```
SH
/opt/powerglove/bin/powerglove-bsb-zap --check \
  --rom "/home/pi/RetroPie/roms/nes/Bad Street Brawler (USA).7z"
```

To make the change, exit any running RetroArch game first:

```
SH
sudo /opt/powerglove/bin/powerglove-bsb-zap --apply \
  --rom "/home/pi/RetroPie/roms/nas/Bad Street Brawler (USA).7z"
```

Without `--apply`, the command is read-only. Exit code 0 means the checked configuration is ready; 2 means action is required. The helper checks the ROM, installed RetroArch and FCEUmm files, NES default and current/legacy per-game emulator selections, and explicit disabling of automatic game options. If FCEUmm is missing, install `lr-fceumm` through RetroPie Setup. If another core is selected, choose `lr-fceumm` for this ROM in the launch menu. The helper does not install emulators or change emulator selections.

Existing game options are preserved. A new file copies the first available options set in game, content-folder, core, then global order; only FCEUmm entries are copied from the global file. Only `fceumm_up_down_allowed` is changed. Updates are atomic, preserve existing file ownership/permissions, and create a `powerglove-bsb-backup-*` directory beside the options file with `RESTORE.txt`. Follow that file to restore the previous options or remove a newly created file. Running the helper again when enabled makes no changes or extra backups.

The default layout is `/opt/retropie`. `--prefix` supports a different installation root, and `--options-dir` specifies a different FCEUmm options directory. Custom core-option path redirects are rejected for manual review. This checks standard RetroPie files; it does not launch the core, resolve arbitrary custom launch scripts/includes, or prove the game loaded the option. Relaunch Bad Street Brawler, confirm its game-specific options are loaded, then test Glove Zap and normal movement. [RetroArch documents game options as complete option sets](#), which is why the helper preserves the inherited settings instead of writing only one option.

Matrix startup hourglass

After the system boot display, the Arduino sketch draws a pulsing hourglass before connecting to Router Bridge. Its display task runs independently of the bridge connection and Python imports, so a slow app startup does not leave the matrix blank once the sketch is running. Loading uses five frames, 220 ms each. The supervisor requests loading before importing the web controls; its normal status updates then select the glove attract animation, profile, or Academy mode. This is an activity indicator, not a percentage or a guarantee of successful startup. If the app never connects, the hourglass continues.

The earlier protected boot display and any interval before the sketch starts belong to the board's system software. This change does not replace those. The display task uses the current Arduino sketch platform's thread API; if its stack cannot be allocated, the initial hourglass remains visible and ordinary loop animation resumes after bridge initialization. Verify a cold boot on the physical matrix, especially the transition from the system display to the hourglass.

Early-start host helper

The helper runs on every boot as the Arduino user, separately from App Lab. It reduces the blank interval after the system animation by releasing the installed sketch early. It preserves the existing hourglass and uses no reset, halt, flash, or replacement of Arduino system services. The one-boot trial was confirmed on the physical board before enabling repeated startup.

The UNO installer performs this automatically. For manual repair, install `scripts/uno-q-early-start.py` at `~/.local/lib/powerglove/uno-q-early-start.py` and `uno-q/powerglove-early-start.service` at `~/.config/systemd/user/powerglove-early-start.service`. The Arduino user must have lingering enabled. Run `systemctl --user daemon-reload` and `systemctl --user enable powerglove-early-start.service`. Disable the old `powerglove-early-start-trial.service` if present. No armed marker is required.

The service waits up to 30 seconds for the router and permits 20 seconds for the debug check and release. It verifies the UNO Q, PowerGlove startup app, Wait for App image header, and four 64-byte code samples. This is not a full integrity check. If the image is unavailable or differs, or the debug pins are busy, it fails without releasing the sketch; normal App Lab startup continues. App Lab may subsequently reset the sketch during its ordinary upload.

Inspect `journalctl --user -b -u powerglove-early-start.service`; disable with `systemctl --user disable powerglove-early-start.service`. Review compatibility after platform updates: the tested loader is Arduino platform 1.0.0 with App Lab 0.13.0. Full installation and removal details are in [Early sketch startup](#).

Matrix animation timing reference

These implementation details support maintenance and physical display checks. For display meanings and user actions, use the [Matrix display guide](#).

Display	Timing and implementation
Startup hourglass	Five frames, 220 ms per frame; repeats while loading
Academy L / tuning T	Eight scan frames, 160 ms per frame, with trailing glow
Tracked profile	Alternates bright and dim every 360 ms
Error X	Alternates X and blank every 420 ms
Pairing	Nine steps of about 650 ms; ID, seven hexadecimal characters in groups of two with one final character, PN, then three digit pairs

The idle glove sequence draws an energy streak, brings the cuff and glove into view, curls and reopens the fingers, moves a spark along the glove, then pulses and pauses before repeating. Pairing temporarily takes display priority, normally for two minutes; a shutdown/off request can clear it sooner. Status precedence can keep a mode letter or idle animation visible instead of X.

Manual installation maintenance

UNO Q updates over Wi-Fi

Use this route after the initial App Lab installation. It updates the Linux application; matrix firmware changes still need **Run** in App Lab.

Set up SSH key access once

- 1 On your development computer, check for an existing public key in `~/.ssh/`. Use only a file ending in `.pub`; never copy its matching private key.
- 2 If you do not have a key, run `ssh-keygen -t ed25519`. Accept the suggested location only if it does not replace an existing key, and follow the passphrase prompts.
- 3 Open your public-key file and copy its complete single line. For the default key, run `cat ~/.ssh/id_ed25519.pub`.
- 4 Connect with `ssh arduino@UNO-Q-NAME.local`. On the UNO Q, run `install -d -m 0700 ~/.ssh`, then `nano ~/.ssh/authorized_keys`.

- 5 Add the public key on a new line, preserving any existing keys. Save with Ctrl+O, confirm the name, and exit with Ctrl+X.
- 6 Run `chmod 0600 ~/.ssh/authorized_keys`, then `exit` to return to your computer. If your private key has a passphrase, make it available through your computer's SSH agent before the unattended deployment check.
- 7 Run `ssh -o BatchMode=yes arduino@UNO-Q-NAME.local hostname`. Continue only when it prints the UNO Q hostname without requesting a login password.

Update the application

- 1 On your development computer, open your project checkout and review local changes with `git status --short`.
- 2 If you are updating from GitHub, run `git pull --ff-only`. Resolve any reported local-change or branch conflict before deploying. Keep the version compatible with the RetroPie installation.
- 3 Run the deployment command below. It preserves private `data/`, restarts the application, and checks its web pages.
- 4 Open Dashboard and Glove Academy to confirm the updated app works. If you changed the matrix sketch, also rebuild and run it through App Lab.

```
SH
scripts/deploy-uno-q-wifi.sh arduino@UNO-Q-NAME.local
```

Use the board's current IP address if its local name is temporarily unavailable. If SSH access fails, reconnect through App Lab before retrying deployment. For local-name failures, see [hostname resolution](#).

Repair or update the shutdown helper

The helper requests `systemctl --no-block halt`, not `poweroff`: the UNO Q can reboot after a power-off request. A `halt` requests that Linux stop; it does not disconnect electrical power. LEDs may remain lit. **Known limitation, confirmed September 3, 2026:** our UNO Q restarted after reaching the halt target both through a powered USB-C hub and when connected directly to a Mac. Shutdown is therefore not a verified way to keep this board stopped. Do not treat the website disappearing as a safe-to-unplug indicator or rely on a fixed countdown. See [Arduino shutdown guidance](#).

The App Lab container is intentionally unprivileged and cannot halt the Linux host. The machine installer already installs the fixed-purpose systemd path helper. To repair or update it separately, run this command from your development computer's project checkout:

```
SH
scripts/install-uno-q-shutdown-helper.sh arduino@UNO-Q-NAME.local
```

Enter the UNO Q `arduino` account password at the remote `sudo` prompt. The script does not read or store it. The helper watches only the fixed `data/shutdown-request` path and can perform only a system halt. After it is installed, **Shutdown** is available on Dashboard and Setup. Each press requires browser confirmation and warns that the UNO Q may restart automatically and that a disconnected website does not confirm it is safe to remove power. Its boot-time `tmpfiles` rule recreates the readiness marker if the UNO Q reboots or App Lab replaces the application directory.

Verify the helper without triggering shutdown:

```
SH
ssh arduino@UNO-Q-NAME.local
systemctl is-enabled powerglove-system-shutdown.path
systemctl is-active powerglove-system-shutdown.path
exit
```

Expected results are [enabled](#) and [active](#). Checks on the cabinet on September 3, 2026 confirmed that the watcher was enabled and active, the readiness marker existed, and both pages displayed the Shutdown button. Requests without confirmation were rejected. Do not test the accepted API path unless you intend to shut down the UNO Q.

RetroPie updates

- 1 On RetroPie, back up customized files under [/etc/powerglove/](#), especially [games.json](#) and [launcher.json](#), using your normal private backup method.
- 2 Open the original source checkout, normally [~/PowerGlove-Vision](#). The installed copy under [/opt/powerglove-src](#) is not a Git checkout.
- 3 Run the commands below. Review [git status --short](#) before pulling; if Git reports a conflict, resolve it before running the installer.
- 4 Resolve any **FAIL** in the installer report, then launch a registered game and check its profile and controls. The installer preserves existing settings and tokens.

```
SH
cd ~/PowerGlove-Vision
git status --short
git pull --ff-only
sudo python3 scripts/setup-machine.py retroPie --peer UNO-Q-NAME.local
```

Duplicate App Lab entries

Importing a newer ZIP may create a timestamped copy. The supported host installer and shutdown helper require [/home/arduino/ArduinoApps/powerglove-vision](#). Keep the working application at that path; do not run host setup from a duplicate. Use the Wi-Fi update procedure for routine Linux application changes. Before removing any duplicate in App Lab, confirm which copy has your private settings and keep only the intended application set to start at boot.

Installation troubleshooting commands

Matrix shows a blinking X

- Confirm that an active gesture profile is selected. **Gestures off** should display the animated glove attract sequence, never the error X.
- Confirm the camera is connected through the powered hub.
- Try another hub port or USB cable.
- Check whether Linux sees a USB camera; internal [qcom-venus-encoder](#) and [qcom-venus-decoder](#) nodes are codecs, not your camera.
- Restart the app after checking power and cabling.

Camera appears only after reconnecting it

This usually indicates USB enumeration or power trouble. Keep the powered hub energized before starting the UNO Q, try another cable, and avoid passive adapters. The app first retries an ordinary camera reopen. With the standard host helper installed, a camera that remains absent for 15 seconds receives one guarded hub-reset attempt; USB Ethernet can disconnect briefly. A camera still absent afterward needs its physical connection checked.

First installation takes several minutes

The first launch downloads and prepares the software it needs. Keep the UNO Q connected to the internet and watch App Lab for progress. Later launches reuse the installed software and normally start faster.

Setup page does not open

- Ordinary settings: <http://UNO-Q-NAME.local:8088/setup>
- Secure pairing: <https://UNO-Q-NAME.local:8443/setup>
- Try the board's IP address if `.local` does not resolve.
- HTTPS and HTTP are not interchangeable on these ports.

Camera is slow to start or missing after reboot

The startup timer includes preparation as well as opening the camera. Allow more time just after a reboot and wait for the camera view before calibrating.

If the app reports that the camera is unavailable, run these commands on the UNO Q:

```
SH
lsusb
ls -l /dev/v4l/by-id/
```

Look for your camera in both outputs. If it is absent from the USB list, check its cable and hub connection, then disconnect and reconnect it. Waiting for MediaPipe will not fix a camera that the operating system cannot detect. Keep Camera set to [auto](#) unless you intentionally selected a particular device. For stage timings and further checks, see [Vision startup and timing](#).

Password pairing fails

- Prepare a new attempt and use its new matrix PIN.
- Confirm the RetroPie username and password can log in through SSH.
- The account must be allowed to run `sudo` with that password.
- Prefer the one-time-code method if password SSH is disabled.

Controller does not appear on RetroPie

```
SH
systemctl is-enabled powerglove-receiver.service
systemctl is-enabled powerglove-receiver.timer
sudo systemctl status powerglove-receiver.service
sudo systemctl status powerglove-receiver.timer
sudo journalctl -u powerglove-receiver.service -n 100 --no-pager
ls -l /dev/uinput
```

Allow 45 seconds after boot. Confirm `uinput` is loaded and `/etc/powerglove/token` is not empty. Expected boot enablement is `disabled` for the service and `enabled` for the timer. The virtual controller appears only after an authenticated packet arrives.

Frontend slowdown or USB-device conflicts

Confirm the receiver service was not enabled directly. Stop it, restart EmulationStation, and start the receiver afterward as an A/B test. If the frontend becomes responsive, restore the supplied timer. Also ensure Pixelcade's `game-select` and `system-select` directories contain only one executable hook each if you use a BitPixel display; executable backup scripts are additional hooks.

Controller exists but does not move

- Select **Start controller** on Setup or Debug.
- Confirm a calibrated hand and controller output on Debug.
- Verify the receiver address and UDP 55355 connectivity.
- Check whether an existing cabinet input merger filters the virtual device.

Profiles do not change

A **profile queued** message confirms authentication and queue admission. Wait for Dashboard to show the new profile; camera startup may still be in progress. For timeouts, check that the UNO Q publishes UDP 55356 through its profile relay. Follow [Check a queued profile change](#) for the command and recovery steps. Check the exact ROM filename, including its archive extension, if the selected profile is **off**.

- Test `powerglove-profile` manually.
- Check `uno_q` and `token_file` in `/etc/powerglove/launcher.json`.
- Confirm both runcommand hooks call the supplied helper scripts.
- Match the exact ROM basename in `/etc/powerglove/games.json`.

FAQ: What if the console name cannot be resolved?

- 1 In **Connection**, enter your console's actual hostname, such as `RETROPIE-NAME.local`, then select **Test console name**. Use a hostname or IPv4 address, not `http://`, a port, or a page path. This tests resolution from the UNO Q app; successful lookup on your laptop alone is not sufficient.
- 2 Confirm the RetroPie console is powered on and connected to your LAN. On its terminal, run `hostname` and `hostname -I` to confirm its name and current addresses. Do not assume an old DHCP address is still correct.

- 3 From the PowerGlove source directory on RetroPie, run `sudo python3 scripts/setup-machine.py retropie --check`. Check Avahi with `systemctl is-active avahi-daemon` and `systemctl is-enabled avahi-daemon`. If setup is incomplete, rerun `sudo python3 scripts/setup-machine.py retropie --peer UNO-Q-NAME.local`, using your board's actual name, and review every FAIL or ACTION result.
- 4 On the UNO Q, from the app directory, run `sudo python3 scripts/setup-machine.py uno-q --check`. This checks the configured destination from inside the application. If installation is incomplete, rerun `sudo python3 scripts/setup-machine.py uno-q`. Do not manually patch `.cache/app-compose.yaml`: App Lab regenerates it. The shipped resolver brick supplies the persistent configuration.
- 5 Check that both machines are on a network that allows communication between devices. Guest Wi-Fi, client isolation, VPN routing, separate VLANs, or multicast filtering can prevent `.local` discovery. mDNS uses UDP port 5353; do not disable your firewall wholesale or expose the app to the Internet to fix discovery.
- 6 As a diagnostic or fallback, enter RetroPie's current LAN IPv4 address in **Connection** and test again. If that works while the name fails, investigate mDNS. For continued use, reserve that address in your router so DHCP does not change it. Save the intended destination using the normal Connection workflow; changing the address does not replace pairing credentials. If RetroPie also contacts the UNO Q by name, check that reverse direction separately.
- 7 If neither name nor IP works, investigate connectivity and the service itself, not just Avahi. A successful name test only establishes name resolution; pairing, the receiver, controller output, and emulator mappings must also work. Retry after boot has finished, then collect the exact error and setup-check results if it still fails. Never share tokens, passwords, or private SSH keys.

After fixing the problem, reboot both machines and repeat **Test console name** before testing gameplay. The app-owned resolver has been verified across a UNO Q reboot and a changed RetroPie DHCP address; no fixed IP entry is required for `.local` use.

Removing the installation

On RetroPie:

```
SH
sudo systemctl disable --now powerglove-receiver.timer
sudo systemctl disable --now powerglove-receiver.service
sudo rm /etc/systemd/system/powerglove-receiver.timer \
/etc/systemd/system/powerglove-receiver.service
sudo systemctl daemon-reload
```

Remove only the PowerGlove lines from the runcommand hooks. After backing up custom profiles, `/opt/powerglove`, `/opt/powerglove-src`, and `/etc/powerglove` may be removed manually.

On the UNO Q, stop the app, disable **Run at startup**, and remove it through Arduino App Lab. Its private `data` directory contains the device token and cached runtime.

Remove the host shutdown helper separately:

```
SH
sudo systemctl disable --now powerglove-system-shutdown.path
sudo rm /etc/systemd/system/powerglove-system-shutdown.path \
/etc/systemd/system/powerglove-system-shutdown.service \
/etc/tmpfiles.d/powerglove-system-shutdown.conf
rm -f /home/arduino/ArduinoApps/powerglove-vision/data/.shutdown-enabled
sudo systemctl daemon-reload
```

Build and install matrix firmware

The verified sketch profile is `arduino:zephyr:unoq` with Arduino Zephyr platform **1.0.0**. Keep the complete pinned `sketch/sketch.yaml`; installing a newer platform by itself does not rebuild or flash PowerGlove. A compile-only check builds the sketch but does not change the running firmware.

From the application directory on the UNO Q, validate the pinned configuration:

```
SH
arduino-cli compile --profile default sketch
```

After a successful compile, use **Run** in App Lab to build and upload the sketch, or restart the app through App Lab's command-line tool:

```
SH
arduino-app-cli app restart /home/arduino/ArduinoApps/powerglove-vision
```

The Wi-Fi deployment script updates Linux files and restarts containers; it does not flash the matrix sketch. Keep a backup before firmware updates. After upload, check Dashboard, ordinary Glove Academy's scanning L, Tune's scanning T, and normal controller operation. Preserve the installed `data/` directory and verify that personal tuning and neutral calibration remain available.

Versioned two-script installation

The normal entry points are `scripts/install-uno-q.sh` and `scripts/install-retropie.sh`. Each resolves a published release tag, downloads its matching package and shared `install-package.py`, verifies SHA-256 checksums, and invokes the shared `setup-machine.py` with `sudo`. Downloads use GitHub HTTPS; checksums detect damaged or mismatched assets, not a compromised release account. The entry scripts themselves must come from the trusted project release. The installation guide uses GitHub's `/releases/latest/download/` asset links and omits version flags. Each script resolves the latest stable release once, then uses that exact tag for the checksum list, shared installer and package. Drafts and prereleases are excluded from automatic selection. A failed `curl` download prevents the chained `bash` command from running; saving the script first also leaves terminal input available for prompts.

To pin a published release, append `--version TAG` to the saved script command, for example `bash install-uno-q.sh --version v0.3.0`. To test a published development prerelease, use `bash install-uno-q.sh --development dev-COMMIT` instead. Replace these example tags with actual published tags, and use the matching option on RetroPie. No GitHub release is created by running an installer.

Option	Behavior
<code>--version TAG</code>	Use one exact published release on both machines. Without a tag, select the latest stable GitHub release.
<code>--development TAG</code>	Explicitly use a published development prerelease, such as <code>dev-COMMIT</code> . This is a release tag, not a branch name.
<code>--peer HOST</code>	UNO Q hostname or IPv4 address for a new RetroPie installation; prompted if omitted in an interactive terminal. Existing destinations are preserved.
<code>--check</code>	Use the installed shared checker. No download, package installation, service restart, or helper release. Sudo may be needed to inspect protected settings.
<code>--help</code>	Show the entry point options without installing.

Supported baseline: an already provisioned UNO Q with App Lab CLI **0.13.0**, Arduino sketch platform **1.0.0**, the shipped pinned libraries, and a Debian-based RetroPie installation with Python **3.7 or newer**. The UNO installer rejects other App Lab CLI versions until compatibility has been validated. It does not upgrade the board OS or App Lab. Allow at least 3 GiB of free space in the Arduino home partition (UNO Q) or `/var/tmp` (RetroPie), and 512 MiB on the system partition for package operations and backups. Larger updates may need more space.

UNO installation stages files in `/home/arduino/ArduinoApps/powerglove-vision`, then uses `arduino-app-cli app start` as the Arduino user to build/upload the sketch and start the app. The root setup phase configures networking, shutdown, default startup, and the user early-start service. It enables Arduino user lingering; it does not issue an early SWD release or reboot during installation. `data/`, generated caches, and the cabinet-only cheat sheet are preserved.

RetroPie installation preserves controller assignments and existing launch hooks. When `~/RetroPie-Setup/retropie_packages.sh` is available for the installing user, setup offers its `install_bin` and `configure` operations for missing RetroArch or FCEUmm. If that setup tool is absent, or its binary installation is unsupported, use RetroPie Setup to prepare the emulator and rerun the installer. No full RetroPie or operating-system installation is attempted.

Registered ROM discovery uses each `/home/*/RetroPie/roms/nnes` directory and the configured game registry. Custom ROM locations need manual game checks. For Bad Street Brawler, setup requests consent before changing emulator selection, then invokes the existing game-specific Glove Zap helper. If no core options yet exist, launch and exit FCEUmm once and rerun. Neither installer supplies ROMs, changes cabinet input mergers, nor adds a new RetroPie shutdown mechanism; existing operating-system shutdown controls remain available.

Updates replace managed files and save replaced versions under `/var/backups/powerglove-vision/TIMESTAMP/`, with a `RESTORE.txt`. Backups created by the package installer are root-only. Resolve failures before continuing; the installer stops rather than claiming success. Rerunning is supported. A failed firmware update may require rerunning the previous release through App Lab's normal sketch upload; restoring Linux files alone does not restore firmware. Never restore backup files wholesale over the filesystem root.

Build and publish installation assets

Generate the public PDFs and App Lab ZIP, then build both installer packages:

```
SH
python3 scripts/build-docs-pdf.py
bash scripts/build-app-lab-package.sh
python3 scripts/build-install-packages.py --version dev-COMMIT
```

`output/install/` contains both ZIPs, the two entry scripts, their shared package installer, and `SHA256SUMS`. Package identity and safe paths are validated both at build time and installation time. Private runtime files are excluded.

The **Build installation release** workflow accepts an existing source revision and a new release tag. Its default builds downloadable workflow artifacts only. Select **Publish a GitHub release** explicitly to make the normal download commands available; leave **prerelease** enabled for feature-branch testing. Use a stable release only after physical fresh-device and upgrade validation. Publishing, tagging, or deploying firmware is not implied by building packages.

Installation validation status

Automated tests cover isolated fresh-directory installs and repeat updates, archive rejection, download failure, and settings preservation. These are not fresh-device tests. The new package installer was also run on the existing RetroPie cabinet: receiver, launch hooks, emulator, and Glove Zap checks passed with no technical failures. The UNO Q full installer still needs an interactive `sudo` run. A provisioned spare UNO Q and RetroPie system, real pairing, live gameplay, and a cold boot are required before declaring the installers validated for release. Run read-only checks on both devices after installation; no automated check proves that a hand gesture controls a game correctly.

Shared installer and application sources

Both installer entry points are generated from `scripts/templates/install.sh.in`. Run `python3 scripts/build-installer-scripts.py` after editing that template; use `--check` to detect drift without changing files.

`scripts/application-payload.py DESTINATION` stages the public application into an empty directory. Both App Lab packaging and Wi-Fi deployment use this selection, excluding private settings, artwork masters, and generated installer archives. `config/profiles.json` is release-owned and is backed up and replaced during updates, so tested shared defaults and new fields take effect consistently. Runtime calibration, pairing, tuning, and cabinet settings remain in place.

Installation ownership manifest

`scripts/installation-manifest.py` manages the application payload in `/home/arduino/ArduinoApps/powerglove-vision` on UNO Q and `/opt/powerglove-src` on RetroPie. Package installation and Wi-Fi deployment use the same implementation. Host service units, launch hooks, controller assignments, and system configuration remain under their existing installers; the payload manifest does not prune them.

Each root contains `.powerglove-install.json` with format version 1, the absolute installation root, release identity, and relative paths with SHA-256 hashes and permission modes. Private data, caches, and the local `docs/cheatsheet.md` are excluded from managed ownership. `config/profiles.json` is deliberately managed and replaced from the release; its previous copy is retained in the update backup.

On the first manifest-enabled update, incoming package paths are installed using the existing backup-and-replace behavior. Unknown files absent from the package are left alone: the installer does not infer an old inventory from the directory. Subsequent updates back up and remove obsolete files only when their bytes and permissions still match the previous manifest. Local changes, including permission changes, are retained and reported, whether or not the new package contains that path. Their old baseline remains recorded so later updates cannot silently adopt or delete them. Restore the previous installed version of a locally modified file before retrying if you want the installer to replace it with the new release.

Updates take an exclusive lock, validate all paths before writing, back up changed and removed files, and publish the new manifest last. A failed write rolls back the payload. A process interruption leaves `.powerglove-install-pending.json`; another update refuses to proceed until recovery. Backups include the old manifest, changed files, `transaction.json`, and `RESTORE.txt`. An installation failure after payload staging (for example, host setup or App Lab startup) does not undo a successfully committed payload; use the reported backup or previous release.

The normal installer `--check` reports missing or modified managed files and an unfinished transaction without changing anything. For payload-only checks:

```
SH
python3 scripts/installation-manifest.py /home/arduino/ArduinoApps/powerglove-vision --check
```

For an interrupted update, stop the application, then use the script from a trusted release staging directory (the installed copy may have been interrupted):

```
SH
sudo python3 scripts/installation-manifest.py /home/arduino/ArduinoApps/powerglove-vision --recover
```

On RetroPie, use `/opt/powerglove-src` and run recovery with `sudo`. Restart and run installer checks afterward. Do not remove the pending journal or manifest to bypass recovery. The manifest never scans or deletes unknown user files or

directories. `--source STAGING --backup BACKUP` applies a staged payload; BACKUP must be a new location outside both source and installation trees. This is used by the deployment script; normal users should use the two standard installers.